BANANA BLASTER — VISUAL APPENDIX

The bones on paper. Every projectile, skill drop, combo, equipment item, set, boss, telegraph, animation, loot table, trait, and formula — with the REAL sprites rendered by the game's own code.

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Counts: 8 drop skills, 28 combos, 8 self-combos, 30 vendor items, 16 traits, 9 named mythic, 39 set items, 20 elemental items, 6 sets, 22 bosses.

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1. Drop Skills (8)

HOW DROP SKILLS WORK (our own system - inspired by Kirby 64): Drop skills drop from enemies as floating pickups. **Walk over** one to pick it up (when slots are unlocked). You have **2 slots** (dual skills). **Two DIFFERENT skills = a Power Combo** (28 pairs, stronger than either single, new effect). **Two of the SAME skill = a self-combo** (8, stronger version of that skill). **Right-click locks the dual slots** (floating skills will NOT be picked up - keeps the combo you like). **Right-click again unlocks** (so you can switch). No grenade - Q does nothing (kyle: "Q throws a grenade makes no sense"). Kirby 64 showed

us two abilities can combine into a new one - that was the inspiration (kyle enjoyed the game). But ours is our own way, and it is SIMPLE: **pick up, right-click to lock, right-click to unlock**. That is the whole system.

Each skill = one sprite + one colour + one overlay silhouette (P1) + one effect.

TRI-FAN `tri`

Sprite: PEA **Colour:** #22e6ff **Size:** 8
Speed: 1x **Overlay:** ring

Effect: 3-banana fan

Combo line:

FROST `ice`

Visual: ICE ORB + BLUE FROST AURA (blue glowing aura over the projectile).

Sprite: ORB **Colour:** #9df3ff **Size:** 10
Speed: 0.9x **Overlay:** halo

Effect: hits slow zombies

Combo line:

BLAZE `fire`

Visual: FLAMETHROWER (procedural flame, bigger, more orange, flickering, points along flight). Not a single pea.

Sprite: WISP **Colour:** #ff8c1a **Size:** 10
Speed: 1.05x **Overlay:** spikes

Effect: burn: dead in 2s

Combo line:

BOOM `bomb`

Sprite: BOMB **Colour:** #ffe14d **Size:** 14
Speed: 0.75x **Overlay:** ring2

Effect: AoE burst on hit

Combo line:

LANCE `pierce`

Sprite: SWORD **Colour:** #ff5df2 **Size:** 12
Speed: 1.25x **Overlay:** spikesLong

Effect: pierces 3 zombies

Combo line:

RAPID `rapid`

Sprite: PEA **Colour:** #7dff4d **Size:** 6
Speed: 1.35x **Overlay:** ringDash

Effect: fire rate +60%

Combo line:

SEEKER `homing`

Sprite: ORB **Colour:** #8f9bff **Size:** 10
Speed: 1x **Overlay:** haloSmall

Effect: bananas home in

Combo line:

SPLIT `split`

Sprite: GEM **Colour:** #ffb13d **Size:** 10
Speed: 1x **Overlay:** spikesDiag

Effect: splits into 2 on hit

Combo line:

2. Combos (28) - Skill Combos

Two skills = one combo. Kirby 64 also has a two-skill combo system - that is where the idea came from. Ours is our own: 8 drop skills, 28 pairs, each pair a new ability. Each pair has its OWN sprite, colour, overlay (both singles' overlays), and ONE distinct twist (P2) on top of the two skills' effects. No two pairs share the same twist.

FIRE PATCHES `bomb+fire`

Visual: FIRE EXPLOSION (bomb + flame, leaves burn patches)

Sprite: BOMB **Colour:** #ff8c1a **Size:** 14
Speed: 0.85x

Twist (P2): AoE x1.5, fire tier 2 (stronger burn)

Effect: BOOM + BLAZE, plus the twist above.

HOMING BOMBS `bomb+homing`

Sprite: BOMB **Colour:** #ffe14d **Size:** 12
Speed: 0.9x

Twist (P2): AoE x1.3, homing turn x1.5

Effect: BOOM + SEEKER, plus the twist above.

FROST BOMBS `bomb+ice`

Visual: BLUE FROST AURA (frosted projectile)

Sprite: BOMB **Colour:** #9df3ff **Size:** 13
Speed: 0.8x

Twist (P2): AoE x1.2, ice tier 2 (freeze)

Effect: BOOM + FROST, plus the twist above.

PIERCING BOMBS `bomb+pierce`

Sprite: SWORD **Colour:** #ffe14d **Size:** 13
Speed: 1x

Twist (P2): AoE x1.3, +1 pierce

Effect: BOOM + LANCE, plus the twist above.

RAPID BOMBS `bomb+rapid`

Sprite: BOMB **Colour:** #ffe14d **Size:** 10
Speed: 1.1x

Twist (P2): AoE x1.3, speed x1.2

Effect: BOOM + RAPID, plus the twist above.

EXPLODING SPLITS `bomb+split`

Sprite: GEM **Colour:** #ffe14d **Size:** 12
Speed: 0.85x

Twist (P2): AoE x1.3, +1 split

Effect: BOOM + SPLIT, plus the twist above.

BOMB FAN `bomb+tri`

Sprite: BOMB **Colour:** #ffe14d **Size:** 11
Speed: 0.85x

Twist (P2): AoE x1.3, +1 fan pea

Effect: BOOM + TRI-FAN, plus the twist above.

HOMING FIRE `fire+homing`

Visual: FLAMETHROWER AUTO-AIMS (homing flame)

Sprite: WISP **Colour:** #ff8c1a **Size:** 10
Speed: 1.05x

Twist (P2): fire tier 2 (stronger burn), homing turn x1.5

Effect: BLAZE + SEEKER, plus the twist above.

THERMAL SHOCK `fire+ice`

Visual: FLAMETHROWER + ICE SPLINTER (kill -> ice shards burst, small dmg) | BLUE + ORANGE AURAS (frost + flame)

Splinter: Kill with the flamethrower (fire+ice) -> enemy splinters into ice shards (70 radius, 1 dmg to nearby enemies).

Sprite: WISP **Colour:** #ff8c1a **Size:** 11
Speed: 1x

Twist (P2): DMG x2

Effect: BLAZE + FROST, plus the twist above.

PIERCING FIRE `fire+pierce`

Visual: FLAMETHROWER PASSES THROUGH (piercing flame)

Sprite: SWORD **Colour:** #ff8c1a **Size:** 12 **Speed:** 1.15x

Twist (P2): +1 pierce, fire tier 2 (stronger burn)

Effect: BLAZE + LANCE, plus the twist above.

RAPID FIRE `fire+rapid`

Visual: FAST FIREBALLS (rapid flames, all burn)

Sprite: WISP **Colour:** #ff8c1a **Size:** 7 **Speed:** 1.3x

Twist (P2): speed x1.2, fire tier 2 (stronger burn)

Effect: BLAZE + RAPID, plus the twist above.

SPLITTING FIRE `fire+split`

Visual: FIREBALLS SCATTER (split shards are fireballs)

Sprite: GEM **Colour:** #ff8c1a **Size:** 11 **Speed:** 1x

Twist (P2): +1 split, fire tier 2 (stronger burn)

Effect: BLAZE + SPLIT, plus the twist above.

FIRE FAN `fire+tri`

Visual: THREE FLAMETHROWERS (fan of 3 flames)

Sprite: WISP **Colour:** #ff8c1a **Size:** 9
Speed: 1.05x

Twist (P2): +1 fan pea, fire tier 2 (stronger burn)

Effect: BLAZE + TRI-FAN, plus the twist above.

HOMING FROST `homing+ice`

Visual: BLUE FROST AURA (frosted projectile)

Sprite: ORB **Colour:** #9df3ff **Size:** 10
Speed: 1x

Twist (P2): ice tier 2 (freeze), homing turn x1.5

Effect: SEEKER + FROST, plus the twist above.

HOMING LANCES `homing+pierce`

Sprite: SWORD **Colour:** #ff5df2 **Size:** 12
Speed: 1.1x

Twist (P2): +1 pierce, homing turn x1.5

Effect: SEEKER + LANCE, plus the twist above.

RAPID HOMING `homing+rapid`

Sprite: PEA **Colour:** #ff5df2 **Size:** 7
Speed: 1.25x

Twist (P2): speed x1.2, homing turn x1.5

Effect: SEEKER + RAPID, plus the twist above.

HOMING SPLITS `homing+split`

Sprite: GEM **Colour:** #ffb13d **Size:** 11
Speed: 1x

Twist (P2): +1 split, homing turn x1.5

Effect: SEEKER + SPLIT, plus the twist above.

HOMING FAN `homing+tri`

Sprite: ORB **Colour:** #ff5df2 **Size:** 9
Speed: 1.05x

Twist (P2): +1 fan pea, homing turn x1.5

Effect: SEEKER + TRI-FAN, plus the twist above.

PIERCING FROST `ice+pierce`

Visual: BLUE FROST AURA (frosted projectile)

Sprite: SWORD **Colour:** #9df3ff **Size:** 12
Speed: 1.1x

Twist (P2): +1 pierce, ice tier 2 (freeze)

Effect: FROST + LANCE, plus the twist above.

RAPID FROST `ice+rapid`

Visual: BLUE FROST AURA (frosted projectile)

Sprite: PEA **Colour:** #9df3ff **Size:** 7
Speed: 1.25x

Twist (P2): speed x1.2, ice tier 2 (freeze)

Effect: FROST + RAPID, plus the twist above.

SPLITTING FROST `ice+split`

Visual: BLUE FROST AURA (frosted projectile)

Sprite: GEM **Colour:** #9df3ff **Size:** 11
Speed: 0.95x

Twist (P2): +1 split

Effect: FROST + SPLIT, plus the twist above.

FROST FAN `ice+tri`

Visual: BLUE FROST AURA (frosted projectile)

Sprite: ORB **Colour:** #9df3ff **Size:** 9
Speed: 0.95x

Twist (P2): +1 fan pea, ice tier 2 (freeze)

Effect: FROST + TRI-FAN, plus the twist above.

RAPID LANCES `pierce+rapid`

Sprite: SWORD **Colour:** #ff5df2 **Size:** 10
Speed: 1.3x

Twist (P2): +1 pierce, speed x1.2

Effect: LANCE + RAPID, plus the twist above.

SPLITTING LANCES `pierce+split`

Sprite: SWORD **Colour:** #ffb13d **Size:** 12
Speed: 1.1x

Twist (P2): +1 pierce, +1 split

Effect: LANCE + SPLIT, plus the twist above.

PIERCING FAN `pierce+tri`

Sprite: SWORD **Colour:** #ff5df2 **Size:** 10
Speed: 1.15x

Twist (P2): +1 pierce, +1 fan pea

Effect: LANCE + TRI-FAN, plus the twist above.

RAPID SPLITS `rapid+split`

Sprite: GEM **Colour:** #ffb13d **Size:** 8
Speed: 1.2x

Twist (P2): +1 split, speed x1.2

Effect: RAPID + SPLIT, plus the twist above.

RAPID FAN `rapid+tri`

Sprite: PEA **Colour:** #ff5df2 **Size:** 7
Speed: 1.25x

Twist (P2): speed x1.2, +1 fan pea

Effect: RAPID + TRI-FAN, plus the twist above.

MEGA FAN `split+tri`

Sprite: GEM **Colour:** #ffb13d **Size:** 9
Speed: 1x

Twist (P2): +1 split, +1 fan pea

Effect: SPLIT + TRI-FAN, plus the twist above.

3. Self-Combos (8)

Same drop twice = a better version of that skill.

Skill	Self-Combo
BOOM	**MEGA BOOM**
BLAZE	**INFERNO**
SEEKER	**PERFECT SEEKER**
FROST	**DEEP FREEZE**
LANCE	**OMNI LANCE**
RAPID	**HYPER RAPID**
SPLIT	**MEGA SPLIT**
TRI-FAN	**MEGA FAN**

4. Vendor Items

Permanent modifiers sold by Vek the vendor. Each has a cost, a max stack, an icon, and a colour.

Icon	Name	Cost	Max	Effect
·	**SMALL BODY**	90	2	hitbox -30% (min 50%)
■	**IRON SKIN**	120	2	-25% damage taken
♥	**HEARTY MEAL**	100	5	+2 max HP, heal 2
↻	**SECOND WIND**	150	1	revive at 3 HP once
○	**BANANA SHIELD**	140	1	bubble blocks 1 hit, 20s CD
V	**VAMPIRE**	90	3	heal per 10 kills
≈	**PHASE BOOTS**	100	1	walk thru cracks
▒	**THICK SKIN**	60	2	

				burn ticks slower, -10% fire dmg
>	**SWIFT BOOTS**	70	3	+15% move speed
»	**DASH BOOTS**	130	1	double-tap dir to dash, 2s CD
=	**STICKY FEET**	80	1	immune: slow zones, +10% speed in them
F	**FIREPROOF**	80	1	immune: burn zones
X	**BOSS BANE**	110	2	+50% dmg to bosses
@	**DEMOLITION**	100	2	explosions +1 damage
→	**LONGER LANCE**	70	3	projectiles pierce +1
○	**BIG BOOM**	80	3	+20% blast radius
»	**SWIFT BANANAS**	60	3	+15% projectile speed
≡	**SWIFT HANDS**	70	3	-10% fire cooldown
!	**ADRENALINE**	80	2	+15% fire rate below half HP
%	**OVERCHARGE**	70	2	gun charges faster
*	**FROSTPROOF**	60	2	slows last +1s, slowed foes 10% slower
♨	**PYRO**	60	2	burns last +1s, burn spreads to 1 nearby
>	**QUICK FEET**	60	2	immune to chill/slow
ϟ	**CHAIN LIGHTNING**	110	2	hits 10% chance to zap nearby foe for 1
?	**LUCKY CHARM**	50	3	+5% skill drop chance, kills drop bonus gems
$	**GREEDY**	50	3	+25% gem value
$	**JACKPOT**	120	1	pay 120, get 150
◷	**TIME WARP**	70	2	enemies spawn 20% slower
✚	**MEDIC**	60	3	hearts heal +1 HP, +3% heart chance
◎	**SCOPE**	60	2	+50% homing turn

5. Gear System: Rarities, Traits, Slots

Rarities

Traits (16)

Every trait is FUN + VISIBLE on its own. Pierce, homing, frost, and fire traits work WITHOUT the matching drop skill (the skill makes them stronger, not required). Previously they were no-ops without the skill.

Rarity	Colour	Affix count

Key	Name	Suffix	Stat	Value	Description
common	#cccccc	1			
uncommon	#4d9fff	2			
rare	#ffe14d	3			
unique	#ff8c1a	4			
mythic	#ff3df2	5			
set	#37ff8b	5			

Traits (prefix / suffix affixes)

Common = 1 affix (prefix OR suffix). Uncommon = 1 prefix + 1 suffix = 2 affixes. Rare = 3 affixes. Unique = 4. Mythic = 5. Set = 5. Name: “Prefix Base of the Suffix”.

Trait	Prefix	Suffix	Stat	Value	
charge	Charged	Storm	gun	0.15	+15% charge rate
fire	Fiery	Phoenix	pyro	30	hits burn enemies (works without BLAZE)
fleet	Fleet	Cheetah	move	0.08	move 8% faster
frost	Frosty	Winter	frost	30	hits slow enemies (works without FROST)
greedy	Greedy	Dragon	greedy	0.1	+10% gem value
heavy	Heavy	Giant	aoe	0.15	+15% blast radius
iron	Ironclad	Mountain	iron	0.08	take 8% less damage
lucky	Lucky	Fortune	lucky	3	+3% drop chance

mend	Mending	Healer	mending	0.5	regenerate 0.5 HP/s
pierce	Piercing	Lance	pierce	1	shots pierce +1 enemy (works without LANCE)
quick	Quick	Falcon	bspd	0.08	+8% pea speed
seek	Seeking	Hunter	homing	0.2	peas home toward enemies (works without SEEKER)
sharp	Sharp	Blade	bane	0.2	+20% damage to bosses
sturdy	Sturdy	Bear	maxhp	1	+1 max HP
swift	Swift	Wind	rate	0.08	attack 8% faster
vamp	Vampiric	Vampire	vamp	0.1	heal 10% of damage dealt

Deterministic traits (named / set items)

LOCKED RULE: the same named / set item ALWAYS has the same traits (hand-written, thematic to the set / item). Random padding is REMOVED for named / set items only (enforceAffixCounts skips unique / mythic / set). Ordinary (common / uncommon / rare) items are still procedurally generated with random affixes. Themes: Sunfire items carry fire, Frostbound carry frost, Moonfang carry swift / sharp / vamp / quick, Venom carry swift / sharp / vamp / pierce, Tempest carry charge / swift, Void carry void-themed traits.

Slots

3 equipment slots: **LEFT HAND, RIGHT HAND, NECKLACE**. Only equipped items grant stats/traits/set bonuses. 2 drop-skill slots. 6 inventory slots (stash, MAX_STASH=6). Stash items grant NOTHING until equipped. Pickup magnet/AoE is FIXED and can NEVER be changed by any item, skill, trait, set bonus, or vendor purchase.

Auto-Weapons: equipped items attack on their own

RULE: With NO equipment, the player only has left-click drop skills (bananas, aimed with the mouse). When an item is EQUIPPED, the player ALSO gets that item's weapon attack - AUTOMATIC, NOT controlled by the player. You control the drop skill projectiles (bananas). You do NOT control the weapon projectiles (they fire on their own timers). That way the player has more projectiles to shoot out, without more buttons.

DAMAGE: Weapon projectile damage = rarity base x type damage multiplier x variance. Rarity base: common=1, uncommon=1, rare=2, unique=2, mythic=3, set=3. Variance: rolled ONCE at drop (0.85x-1.15x, +/-15%) and stored on the item, so two copies of the same weapon can hit slightly differently. Example (nominal variance 1.00): a mythic sword = 3 x 1.5 x 1.00 = 4.5 dmg (3.83-5.18 with variance). A common dagger = 1 x 0.7 x 1.00 = 0.7 dmg. Wave-found does NOT affect damage (auto weapons get no wave boost - looting items is the point).
SPEED (cooldown between auto-attacks, frames @60fps): common=120 (slowest), uncommon=120 (slowest), rare=100 (a bit faster), unique=80 (even faster), mythic=60 (faster than unique), set=45 (fastest). Rarity makes weapons BOTH stronger AND faster. Then the weapon TYPE modifies it (see table): a sword is slower than a dagger, a dagger is quicker.
SHIELD BUBBLE CD: After the bubble pops (blocks its hits), it goes on cooldown before it comes back. CD by rarity: common/ uncommon=10s, rare=8s, unique=6s, mythic=5s, set=4s. Two shields = the bubble blocks 2 hits, CD = the better (shorter) shield's CD. Bubble does NOT stack with the BANANA SHIELD vendor item (that is a separate bubble with its own 20s CD).
WEAPON TYPES ARE THEMED AROUND THEMSELVES: each weapon type has its own damage + speed multiplier, on top of rarity. A sword does MORE damage than a dagger but is SLOWER. A dagger does LESS damage but is QUICKER. Final damage = rarity base x type damage multiplier x variance (variance rolled once at drop, 0.85x-1.15x, stored on the item). Final cooldown = rarity cooldown x type speed multiplier.

Weapon	Visual	Auto-attack	Damage	Speed
Sword		Wave slash to the FRONT - a glowing crescent slash (like a meteor) with a blue-white trail.	x1.5 (more dmg, slower)	x1.3 (slower)
Dagger		Quick STAB in front - a small stabby thing, happens MORE (quicker) because it is a dagger.	x0.7 (less dmg, quicker)	x0.6 (quicker)
Javelin		Shoots a SMALL JAVELIN out (projectile, medium speed).	x1.2	x1.1
Crossbow		Shoots a CROSSBOLT out - a bit FASTER than the javelin.	x0.8 (weaker, faster)	x0.7 (faster)

Shield	|	BUBBLE instead of attack - a defensive bubble. 1 shield = blocks 1 hit. 2 shields = STRONGER bubble (blocks 2 hits). Bubble has a CD: after it pops, it comes back after the CD.	blocks 1-2 hits	CD by rarity
Wand	|	Magic BOLT - homes toward the nearest enemy.	x1.0 (balanced)	x1.0 (balanced)
Staff	|	AoE BLAST - a small explosion at a nearby enemy.	x1.4 (big, slow)	x1.4 (slower)
Spear	|	Long THRUST - pierces through enemies.	x1.3 (long reach)	x1.2

Necklace (neck): NO auto-attack. A necklace is an accessory, not a weapon - it gives +1 max HP and traits, but NO projectile, NO auto-attack. Only HAND weapons (sword, dagger, javelin, crossbow, wand, staff, spear) get auto-attacks. The shield is also hand - it gets a bubble (not an attack).

Auto-Weapon DAMAGE TABLE (7 attacking weapons, NOMINAL variance 1.00)

7 attacking weapons (sword, dagger, javelin, crossbow, wand, staff, spear). Shield = bubble (no damage). Necklace = NO auto-attack. Damage = rarity base x type damage multiplier x variance. Damage is FRACTIONAL (no rounding) - the game tracks HP as a float (burn, thorns, zone damage are already fractional). Table shows NOMINAL values (variance 1.00). Actual damage = nominal x variance (0.85-1.15, rolled once at drop). Example: a mythic sword = 3 x 1.5 x 1.00 = 4.5 dmg nominal (3.83-5.18 with variance). A common dagger = 1 x 0.7 x 1.00 = 0.7 dmg nominal (0.60-0.81 with variance).

Weapon	Common	Uncommon	Rare	Unique	Mythic	Set
Sword (x1.5)	1.5	1.5	3.0	3.0	4.5	4.5
Dagger (x0.7)	0.7	0.7	1.4	1.4	2.1	2.1
Javelin (x1.2)	1.2	1.2	2.4	2.4	3.6	3.6
Crossbow (x0.8)	0.8	0.8	1.6	1.6	2.4	2.4
Wand (x1.0)	1.0	1.0	2.0	2.0	3.0	3.0
Staff (x1.4)	1.4	1.4	2.8	2.8	4.2	4.2

Spear (x1.3)	1.3	1.3	2.6	2.6	3.9	3.9

Auto-Weapon SPEED TABLE (cooldown between auto-attacks)

Cooldown = rarity cooldown x type speed multiplier. Frames rounded to nearest integer (frames are integers). Seconds = frames / 60. Lower = faster. A sword is slower than a dagger; a dagger is quicker. Shield has NO attack cooldown (bubble CD table is separate: 10/8/6/5/4s).

Weapon	Common	Uncommon	Rare	Unique	Mythic	Set
Sword (x1.3)	156 frames (2.60s)	156 frames (2.60s)	130 frames (2.17s)	104 frames (1.73s)	78 frames (1.30s)	58 frames (0.97s)
Dagger (x0.6)	72 frames (1.20s)	72 frames (1.20s)	60 frames (1.00s)	48 frames (0.80s)	36 frames (0.60s)	27 frames (0.45s)
Javelin (x1.1)	132 frames (2.20s)	132 frames (2.20s)	110 frames (1.83s)	88 frames (1.47s)	66 frames (1.10s)	50 frames (0.83s)
Crossbow (x0.7)	84 frames (1.40s)	84 frames (1.40s)	70 frames (1.17s)	56 frames (0.93s)	42 frames (0.70s)	31 frames (0.52s)
Wand (x1.0)	120 frames (2.00s)	120 frames (2.00s)	100 frames (1.67s)	80 frames (1.33s)	60 frames (1.00s)	45 frames (0.75s)
Staff (x1.4)	168 frames (2.80s)	168 frames (2.80s)	140 frames (2.33s)	112 frames (1.87s)	84 frames (1.40s)	63 frames (1.05s)
Spear (x1.2)	144 frames (2.40s)	144 frames (2.40s)	120 frames (2.00s)	96 frames (1.60s)	72 frames (1.20s)	54 frames (0.90s)

Base names (procedural gear)

Common / uncommon / rare gear is procedurally generated: a base name (weapon or necklace) + affixes from the trait table above. The bases:

Weapons (hand): Sword (/), Dagger (;), Wand (*), Shield (]), Staff (|), Spear (^), Javelin (!), Crossbow (+). Each weapon type has its OWN sprite. Base stats: sword +5% bullet speed, dagger +5% fire rate, wand +5% homing, shield +5% armor, staff +5% AoE, spear +5% pierce, javelin +8% bullet speed, crossbow +8% fire rate.

Necklaces (neck, icon o, base +1 max HP): String Necklace, Bone Charm, Copper Pendant, Wooden Beads, Old Locket, Stone Charm, Rope Necklace, Clay Pendant.

6. Named Gear: Mythic, Set, Elemental

P6: Every named item has ONE distinct effect (it.effect). Mythic: Worldsplitter=shockwave (kill -> small AoE), Aegis of Dawn=dawnshield (once per wave, negate one hit), Voidtouched Dagger=LINE TRAP (IMPLEMENTED, see §20.6): CD 2s; every 2s, next shot leaves a line; line lasts 3s (180f); each monster crossing takes 4 dmg (once per line); 1 line at a time (new replaces old), Stormcaller Wand=stormcall (every 10s, lightning strikes a random enemy), Heart of the Mountain=thorns (attacker takes 1 dmg), World Staff=knockback (hits knock back), Worldpiercer=worldpierce (shots pierce everything), Skybreaker=SUMMON A PERIODIC STORM (IMPLEMENTED, see §20.5): every 480 frames (8s) a storm strikes the arena (lightning + rain, 120 px radius, 3 dmg per 30f tick, 4 ticks = 12 total), Doombringer=doom (kill -> nearby enemies take 1 dmg). Unique: Fallen Blade=firstblood (+1 dmg on first hit), Cinder Crown=burnaura (nearby enemies burn), Glacial Shard=chillshot (shots slow), Baton of Endings=encore (boss kill +1 HP), Honey Dipper=sticky (25% extra gem), Squeak Hammer=squeak (hits knock back), Stinger=sting (shots poison), Tempest Staff=tempest (every 12s, storm AoE), Piercer=pierce1 (shots pierce +1), Sky Javelin=velocity (shots +30% speed), True Aim=trueshot (shots home), Wobble Shield=wobble (take 1 less dmg), Starlight Wand=starhome (shots home strongly), Golden Peel=goldrush (10% goldnug on kill).

Mythic (named, fixed)

Icon	Name	Slot	Base	Traits	Effect	Lore
/	**Worldsplitter**	hand	bspd: 0.15, bane: 0.2	Sharp/ Blade (bane), Swift/Wind (rate), Quick/ Falcon (bspd), Heavy/ Giant (aoe)	shockwave: Kill -> small shockwave (1 dmg AoE)	*Forged where the sky cracked.*
]	**Aegis of Dawn**	hand	iron: 0.15, maxhp: 2	Ironclad/ Mountain (iron), Sturdy/ Bear	dawnshield: Once per wave: negate one hit entirely	*The first light eve blocked.*

				(maxhp), Mending/ Healer (mending)		
;	**Voidtouched Dagger**	hand	rate: 0.15, pierce: 1	Swift/Wind (rate), Sharp/ Blade (bane), Vampiric/ Vampire (vamp), Piercing/ Lance (pierce)	LINE TRAP (IMPLEMENTED): CD 2s; every 2s, next shot leaves a line; line lasts 3s (180f); each monster crossing takes 4 dmg (once per line); 1 line at a time (new replaces old). Lore updated to line-trap (IMPLEMENTED 2026-10-14).	*It cuts a line in t air. Ever 2s, the next shot leaves a line (3s) crossing enemies take 4 dm once per line.*
*	**Stormcaller Wand**	hand	aoe: 0.15, homing: 0.1	Heavy/ Giant (aoe), Seeking/ Hunter (homing), Charged/ Storm (gun), Quick/ Falcon (bspd)	stormcall: Every 10s: lightning strikes a random enemy (1 dmg)	*It pointe at a stor once. The storm apologize*
o	**Heart of the Mountain**	neck	maxhp: 4	Sturdy/ Bear (maxhp), Mending/ Healer (mending), Ironclad/ Mountain (iron), Lucky/ Fortune (lucky)	thorns: Attacker takes 1 dmg (nearest enemy)	*The mountain' pulse, bottled.*
|	**World Staff**	hand	aoe: 0.15, bane: 0.15	Heavy/ Giant (aoe), Sharp/	knockback: Hits knock back (30)	*The world leans on it.*

				Blade (bane), Swift/Wind (rate), Quick/ Falcon (bspd)		
^	**Worldpiercer**	hand	pierce: 1, bane: 0.2	Sharp/ Blade (bane), Swift/Wind (rate), Piercing/ Lance (pierce), Heavy/ Giant (aoe)	worldpierce: Shots pierce everything (infinite)	*Dragons fear it. do knight*
!	**Skybreaker**	hand	bspd: 0.15, rate: 0.1	Swift/Wind (rate), Sharp/ Blade (bane), Charged/ Storm (gun), Quick/ Falcon (bspd)	SUMMON A PERIODIC STORM (IMPLEMENTED, see § 20.5): every 480 frames (8s) a storm strikes (120 px radius, 3 dmg per 30f tick, 4 ticks = 12 total, lightning, cause skystorm).	*Faster th the storm it summon Every 8s storm strikes t arena (2s damages enemies).*
+	**Doombringer**	hand	rate: 0.15, bane: 0.15	Sharp/ Blade (bane), Swift/Wind (rate), Seeking/ Hunter (homing), Heavy/ Giant (aoe)	doom: Kill -> nearby enemies take 1 dmg	*The last thing you'll hear.*

Unique (named, fixed)

Icon	Name	Slot	Base	Traits	Effect	Lore
/	**Fallen Blade**	hand	bspd: 0.1	Sharp/ Blade	firstblood: +1 dmg on	*The blade Ronin*

				(bane), Swift/Wind (rate), Quick/ Falcon (bspd)	first hit per enemy (enemy at full HP)	*dropped. It still remembers duels.*
o	**Cinder Crown**	neck	maxhp: 2	Fiery/ Phoenix (pyro), Sturdy/ Bear (maxhp), Mending/ Healer (mending)	burnaura: Nearby enemies burn (DoT)	*The Pyre Queen's crown. Warm to the touch, forever.*
*	**Glacial Shard**	hand	homing: 0.1	Frosty/ Winter (frost), Quick/ Falcon (bspd), Ironclad/ Mountain (iron)	chillshot: Shots slow enemies	*A splinter of Frostbite's heart.*
*	**Baton of Endings**	hand	aoe: 0.1	Heavy/ Giant (aoe), Swift/Wind (rate), Sharp/ Blade (bane)	encore: Boss kill: +1 HP	*The Conductor's baton. Every end has a score.*
*	**Honey Dipper**	hand	greedy: 0.15	Greedy/ Dragon (greedy), Lucky/ Fortune (lucky), Mending/ Healer (mending)	sticky: 25% extra gem on kill	*Honeybear's dipper. Everything sticks to it, even luck.*
/	**Squeak Hammer**	hand	rate: 0.1	Swift/Wind (rate), Lucky/ Fortune (lucky), Quick/ Falcon (bspd)	squeak: Hits knock back (15)	*The Jester's hammer. Squeaks on impact. Devastating.*

;	**Stinger**	hand	bane: 0.2	Sharp/ Blade (bane), Swift/Wind (rate), Vampiric/ Vampire (vamp)	sting: Shots poison (DoT)	*Queen Vespa's stinger. Do not lick.*
|	**Tempest Staff**	hand	aoe: 0.12	Heavy/ Giant (aoe), Seeking/ Hunter (homing), Charged/ Storm (gun)	tempest: Every 12s: storm AoE at a random enemy (1 dmg)	*The storm's own walking stick.*
^	**Piercer**	hand	pierce: 1	Sharp/ Blade (bane), Swift/Wind (rate), Piercing/ Lance (pierce)	pierce1: Shots pierce +1 enemy	*One thrust, three enemies.*
!	**Sky Javelin**	hand	bspd: 0.12	Swift/Wind (rate), Sharp/ Blade (bane), Quick/ Falcon (bspd)	velocity: Shots +30% speed	*It never lands. It just keeps going.*
+	**True Aim**	hand	rate: 0.12	Sharp/ Blade (bane), Swift/Wind (rate), Seeking/ Hunter (homing)	trueshot: Shots home (level 1)	*Never misses. Never forgives.*
]	**Wobble Shield**	hand	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp)	wobble: Take 1 less damage (min 1)	*The Baron's shield. Wobbles, but never falls.*
*		hand				

	Starlight Wand		homing: 0.15	Seeking/ Hunter (homing), Swift/Wind (rate), Charged/ Storm (gun)	starhome: Shots home strongly (level 2)	*Starweaver dropped a stitch of the night sky.*
o	**Golden Peel**	neck	maxhp: 3, greedy: 0.2	Sturdy/ Bear (maxhp), Greedy/ Dragon (greedy), Lucky/ Fortune (lucky)	goldrush: 10% goldnug on kill	*The Banana King's peel. Royalty is slippery.*
o	**Oracle Gear**	hand	rate: 0.1	Swift/Wind (rate), Quick/ Falcon (bspd), Charged/ Storm (gun)	—	*A gear that turns before it is wound.*
o	**Timepiece**	neck	maxhp: 2	Sturdy/ Bear (maxhp), Mending/ Healer (mending), Lucky/ Fortune (lucky)	—	*It ticks backward. Do not ask why.*
;	**Brass Key**	hand	pierce: 1	Sharp/ Blade (bane), Swift/Wind (rate), Piercing/ Lance (pierce)	—	*Opens any lock. Including time.*
/	**Candy Cane**	hand	bspd: 0.1	Fiery/ Phoenix (pyro), Swift/Wind (rate), Sharp/	—	*Sweet, sharp, and slightly sticky.*

				Blade (bane)		
o	**Gumball**	neck	maxhp: 2	Sturdy/ Bear (maxhp), Greedy/ Dragon (greedy), Mending/ Healer (mending)	—	*Chew slowly. It lasts forever.*
*	**Sugar Wand**	hand	aoe: 0.1	Fiery/ Phoenix (pyro), Seeking/ Hunter (homing), Charged/ Storm (gun)	—	*Waves of pure sugar. Cavities included.*
o	**Thorn Crown**	neck	maxhp: 2	Sturdy/ Bear (maxhp), Mending/ Healer (mending), Ironclad/ Mountain (iron)	—	*Every rose has its price.*
/	**Bramble Blade**	hand	bspd: 0.1	Sharp/ Blade (bane), Swift/Wind (rate), Vampiric/ Vampire (vamp)	—	*It grows sharper with every cut.*
o	**Seed Pouch**	neck	greedy: 0.15	Greedy/ Dragon (greedy), Lucky/ Fortune (lucky), Mending/ Healer (mending)	—	*Plant one. Get a forest.*
]	**Leviathan Scale**	hand	iron: 0.1	Ironclad/ Mountain (iron),	—	

				Sturdy/ Bear (maxhp), Frosty/ Winter (frost)		*Harder than the deep is dark.*
o	**Abyss Pearl**	neck	maxhp: 3	Sturdy/ Bear (maxhp), Mending/ Healer (mending), Lucky/ Fortune (lucky)	—	*It glows with light that never was.*
*	**Tidecaller**	hand	homing: 0.1	Seeking/ Hunter (homing), Swift/Wind (rate), Charged/ Storm (gun)	—	*The tide comes when it calls.*
*	**Prism Core**	hand	aoe: 0.1	Frosty/ Winter (frost), Seeking/ Hunter (homing), Charged/ Storm (gun)	—	*Light goes in. Rainbows come out. Sharp ones.*
;	**Crystal Shard**	hand	bane: 0.2	Frosty/ Winter (frost), Sharp/ Blade (bane), Swift/Wind (rate)	—	*A splinter of the Warden's gaze.*
o	**Warden Lens**	neck	maxhp: 2	Sturdy/ Bear (maxhp), Mending/ Healer (mending), Frosty/ Winter (frost)	—	*See every facet. Including your own.*

|	**Sunfire Staff**	hand	aoe: 0.1	Fiery/ Phoenix (pyro), Heavy/ Giant (aoe), Seeking/ Hunter (homing), Charged/ Storm (gun)	—	*The sun's own scepter.*
^	**Sunfire Spear**	hand	pierce: 1	Fiery/ Phoenix (pyro), Sharp/ Blade (bane), Swift/Wind (rate), Piercing/ Lance (pierce)	—	*Burns through ranks.*
!	**Sunfire Javelin**	hand	bspd: 0.1	Fiery/ Phoenix (pyro), Swift/Wind (rate), Quick/ Falcon (bspd), Charged/ Storm (gun)	—	*A falling star, thrown.*
+	**Sunfire Crossbow**	hand	rate: 0.1	Fiery/ Phoenix (pyro), Sharp/ Blade (bane), Seeking/ Hunter (homing), Heavy/ Giant (aoe)	—	*Bolts of pure dawn.*

Set items (fixed, per set)

Icon	Name	Slot	Set	Base	Traits	Lore

/	**Aegis Blade**	hand	aegis	bspd: 0.1	Ironclad/ Mountain (iron), Sharp/ Blade (bane), Swift/Wind (rate), Sturdy/ Bear (maxhp)	*A blade forged from shield-metal.*
o	**Aegis Pendant**	neck	aegis	maxhp: 2	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Lucky/ Fortune (lucky), Mending/ Healer (mending)	*A miniature shield, worn as a charm.*
]	**Aegis Shield**	hand	aegis	iron: 0.15	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Mending/ Healer (mending)	*The shield that never breaks.*
/	**Crown Blade**	hand	crown	bspd: 0.1	Lucky/ Fortune (lucky), Sharp/ Blade (bane), Greedy/ Dragon (greedy), Swift/Wind (rate)	*A blade fit for a king.*
o	**Crown Jewel**	neck	crown	maxhp: 3	Sturdy/ Bear (maxhp), Lucky/ Fortune (lucky), Greedy/	*The jewel from the Banana King's crown.*

					Dragon (greedy), Mending/ Healer (mending)	
]	**Crown Shield**	hand	crown	iron: 0.1	Sturdy/ Bear (maxhp), Lucky/ Fortune (lucky), Greedy/ Dragon (greedy)	*The royal guard's shield.*
+	**Deadeye**	hand	deadeye	rate: 0.1	Sharp/ Blade (bane), Swift/Wind (rate), Seeking/ Hunter (homing), Heavy/ Giant (aoe)	*Never misses. Never forgives.*
o	**Deadeye Pendant**	neck	deadeye	maxhp: 2	Sturdy/ Bear (maxhp), Seeking/ Hunter (homing), Lucky/ Fortune (lucky), Mending/ Healer (mending)	*The eye that never blinks.*
]	**Deadeye Shield**	hand	deadeye	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Seeking/ Hunter (homing)	*A shield with a crosshair etched in it.*
^	**Dragon Lance**	hand	dragonlance	pierce: 1	Sharp/ Blade (bane), Swift/Wind	*Dragons fear it. So do knights.*

					(rate), Piercing/ Lance (pierce), Heavy/ Giant (aoe)	
o	**Dragon Pendant**	neck	dragonlance	maxhp: 2	Sturdy/ Bear (maxhp), Sharp/ Blade (bane), Lucky/ Fortune (lucky), Mending/ Healer (mending)	*A dragon's tooth, worn as a trophy.*
]	**Dragon Shield**	hand	dragonlance	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Sharp/ Blade (bane)	*Scales of a dragon, shaped into a shield.*
/	**Frostbound Blade**	hand	frostbound	bspd: 0.1	Frosty/ Winter (frost), Swift/Wind (rate), Sharp/ Blade (bane), Ironclad/ Mountain (iron)	*Frostbite's apology. It never melts.*
]	**Glacial Aegis**	hand	frostbound	iron: 0.1	Frosty/ Winter (frost), Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp)	*Winter, held at arm's length.*
o		neck	frostbound		Frosty/ Winter	

	Winter's Heart			maxhp: 2	(frost), Sturdy/ Bear (maxhp), Mending/ Healer (mending), Lucky/ Fortune (lucky)	*It beats once per blizzard.*
;	**Fang of the Moon**	hand	moonfang	rate: 0.1	Swift/Wind (rate), Sharp/ Blade (bane), Vampiric/ Vampire (vamp), Quick/ Falcon (bspd)	*Moonhowl's fang. It howls back.*
o	**Moonstone Collar**	neck	moonfang	maxhp: 2	Sturdy/ Bear (maxhp), Mending/ Healer (mending), Lucky/ Fortune (lucky), Quick/ Falcon (bspd)	*The moon wears one too.*
]	**Pelt of Night**	hand	moonfang	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Frosty/ Winter (frost)	*Woven from midnight itself.*
o	**Star Pendant**	neck	starweaver	maxhp: 2	Sturdy/ Bear (maxhp), Seeking/ Hunter (homing), Lucky/ Fortune	*A star, worn close to the heart.*

					(lucky), Mending/ Healer (mending)	
]	**Star Shield**	hand	starweaver	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Seeking/ Hunter (homing)	*A shield made of captured starlight.*
*	**Star Wand**	hand	starweaver	homing: 0.1	Seeking/ Hunter (homing), Swift/Wind (rate), Charged/ Storm (gun), Quick/ Falcon (bspd)	*Starweaver's own wand. It hums with starlight.*
o	**Storm Pendant**	neck	stormcaller	maxhp: 2	Sturdy/ Bear (maxhp), Heavy/ Giant (aoe), Lucky/ Fortune (lucky), Mending/ Healer (mending)	*Thunder, bottled.*
]	**Storm Shield**	hand	stormcaller	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Heavy/ Giant (aoe)	*It has never been struck by lightning. It IS lightning.*
|	**Storm Staff**	hand	stormcaller	aoe: 0.1	Heavy/ Giant (aoe), Seeking/ Hunter	*The storm's own walking stick.*

					(homing), Charged/ Storm (gun), Quick/ Falcon (bspd)	
]	**Ashen Aegis**	hand	sunfire	iron: 0.1	Fiery/ Phoenix (pyro), Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Mending/ Healer (mending)	*Ash that refused to fall.*
/	**Cinder Blade**	hand	sunfire	bspd: 0.1	Fiery/ Phoenix (pyro), Swift/Wind (rate), Sharp/ Blade (bane), Heavy/ Giant (aoe)	*Quenched in the Pyre Queen's tears.*
o	**Ember Pendant**	neck	sunfire	maxhp: 2	Fiery/ Phoenix (pyro), Sturdy/ Bear (maxhp), Greedy/ Dragon (greedy), Mending/ Healer (mending)	*An ember that never cools.*
]	**Tempest Aegis**	hand	tempest	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Charged/	*The storm's shield. It crackles.*

					Storm (gun)	
/	**Tempest Blade**	hand	tempest	bspd: 0.1	Swift/Wind (rate), Sharp/ Blade (bane), Charged/ Storm (gun), Quick/ Falcon (bspd)	*It hums before the thunder.*
o	**Tempest Heart**	neck	tempest	maxhp: 2	Charged/ Storm (gun), Sturdy/ Bear (maxhp), Lucky/ Fortune (lucky), Mending/ Healer (mending)	*It beats like distant thunder.*
o	**Thunder Pendant**	neck	thunderbolt	maxhp: 2	Sturdy/ Bear (maxhp), Swift/Wind (rate), Lucky/ Fortune (lucky), Mending/ Healer (mending)	*A spark of the storm.*
]	**Thunder Shield**	hand	thunderbolt	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Swift/Wind (rate)	*It rings like thunder when struck.*
!	**Thunderbolt**	hand	thunderbolt	bspd: 0.1	Swift/Wind (rate), Sharp/ Blade (bane), Charged/	*Faster than the storm it summons.*

					Storm (gun), Quick/ Falcon (bspd)	
]	**Swarm Carapace**	hand	venom	iron: 0.1	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Vampiric/ Vampire (vamp)	*Woven from a thousand wings.*
o	**Swarmheart**	neck	venom	maxhp: 2	Vampiric/ Vampire (vamp), Sturdy/ Bear (maxhp), Lucky/ Fortune (lucky), Mending/ Healer (mending)	*It buzzes when danger is near.*
;	**Venomfang**	hand	venom	rate: 0.1	Sharp/ Blade (bane), Swift/Wind (rate), Vampiric/ Vampire (vamp), Piercing/ Lance (pierce)	*Queen Vespa's spare. She has more.*
]	**Void Aegis**	hand	void	iron: 0.1	Vampiric/ Vampire (vamp), Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp), Mending/ Healer (mending)	*It hungers, quietly.*

/	**Void Blade**	hand	void	bspd: 0.1	Vampiric/ Vampire (vamp), Sharp/ Blade (bane), Seeking/ Hunter (homing), Quick/ Falcon (bspd)	*It drinks the light.*
o	**Voidheart**	neck	void	maxhp: 2	Vampiric/ Vampire (vamp), Sturdy/ Bear (maxhp), Mending/ Healer (mending), Lucky/ Fortune (lucky)	*It beats once per void.*

Elemental items (themed, NOT sets)

Icon	Name	Slot	Element	Base	Traits	Lore
/	**Ember Blade**	hand	fire	bspd: 0.08	Fiery/ Phoenix (pyro), Swift/ Wind (rate)	*It smolders even in rain.*
/	**Flamebrand**	hand	fire	bspd: 0.08	Fiery/ Phoenix (pyro), Heavy/ Giant (aoe)	*Forged in a dragon's breath.*
*	**Cinder Wand**	hand	fire	homing: 0.08	Fiery/ Phoenix (pyro), Seeking/ Hunter (homing)	*Sparks fly when you wave it.*
|	**Inferno Staff**	hand	fire	aoe: 0.08	Fiery/ Phoenix (pyro),	

					Heavy/ Giant (aoe)	*The staff that burned a forest.*
/	**Frostbite Blade**	hand	ice	bspd: 0.08	Frosty/ Winter (frost), Swift/ Wind (rate)	*Cold enough to freeze blood.*
^	**Glacier Spear**	hand	ice	pierce: 0.08	Frosty/ Winter (frost), Sharp/ Blade (bane)	*It never thaws.*
*	**Winter Wand**	hand	ice	homing: 0.08	Frosty/ Winter (frost), Seeking/ Hunter (homing)	*Summons snowflakes in summer.*
]	**Arctic Shield**	hand	ice	iron: 0.08	Frosty/ Winter (frost), Ironclad/ Mountain (iron)	*Ice that never cracks.*
/	**Stormcaller Blade**	hand	lightning	bspd: 0.08	Charged/ Storm (gun), Swift/ Wind (rate)	*It hums during thunderstorms.*
!	**Thunder Javelin**	hand	lightning	bspd: 0.1	Charged/ Storm (gun), Sharp/ Blade (bane)	*Faster than the sound it makes.*
*	**Volt Wand**	hand	lightning	homing: 0.08	Charged/ Storm (gun), Seeking/ Hunter (homing)	*Crackles when you hold it.*
+	**Tempest Crossbow**	hand	lightning	rate: 0.1	Charged/ Storm	

					(gun), Sharp/ Blade (bane)	*Every bolt finds its mark.*
/	**Voidreaper**	hand	void	bspd: 0.08	Swift/ Wind (rate), Sharp/ Blade (bane)	*It cuts magic itself. Spells unravel near it.*
;	**Null Blade**	hand	void	rate: 0.08	Swift/ Wind (rate), Vampiric/ Vampire (vamp)	*Where it strikes, magic dies.*
*	**Silencer Wand**	hand	void	homing: 0.08	Seeking/ Hunter (homing), Swift/ Wind (rate)	*It silences spells. Wizards fear it.*
|	**Oblivion Staff**	hand	void	aoe: 0.08	Heavy/ Giant (aoe), Sharp/ Blade (bane)	*It erases what it touches. Even memories.*
/	**Crusher**	hand	physical	bspd: 0.08	Heavy/ Giant (aoe), Sharp/ Blade (bane)	*Simple. Heavy. Effective.*
^	**Ironbreaker**	hand	physical	pierce: 0.08	Heavy/ Giant (aoe), Piercing/ Lance (pierce)	*It breaks iron. And bones.*
|	**Battering Ram**	hand	physical	aoe: 0.08	Heavy/ Giant (aoe), Ironclad/ Mountain (iron)	*A siege weapon, handheld.*
]		hand	physical			

	Anvil Shield			iron: 0.08	Ironclad/ Mountain (iron), Sturdy/ Bear (maxhp)	*It has never been dented.*

7. Sets + Set Skills

3 pieces of a set = transformation + 2 thematic skills + drop skills LOCKED. Set skills are AoE telegraphs (0.5s telegraph, 1–2s cooldown), stronger than any drop-skill combo.

WHOA RULE (locked): wielding the set skills must make people go 'whoa' — the set skills (especially RIGHT clicks) must feel spectacular and powerful (big visuals, big payoff), not just stat bumps.
SHARED SET-SKILL SCALING: damage = skill dmg × skillStarMul (1.25^avgStars). Vs boss: ceil(dmg × 1.5 × (1 + 0.5 × mods.bane + gear.bane)). Each skill's own FORMULA comment in the game source gives its exact numbers (extracted below where present).

aegis

2pc: +15% armor | **3pc:** AEGIS FORM, +40% armor, +1 max HP
(weapon set — bonuses from computeGear)

AEGIS: 2pc +15% armor | 3pc AEGIS FORM (+40% armor, +1 max HP)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	D
	SHIELD BASH	8	95	30 frames	70 frames	aim	knockback	S b A n e K b
	BULWARK	6	140	30 frames	90 frames	player	knockback	B s A y S f

crown

2pc: +10% gem value | **3pc:** CROWN FORM, +25% gem value, +1 max HP, +10 lucky *(weapon set — bonuses from computeGear)*

CROWN: 2pc +10% gold | 3pc CROWN FORM (+25% gold, +1 max HP, +10% lucky)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	Desc
👑	**ROYAL DECREE**	10	100	30 frames	80 frames	aim	—	Roya decr dmg near enem drop gems
✦	**GOLDEN NOVA**	9	135	30 frames	100 frames	player	—	Gold nova AoE you. burs

deadeye

2pc: +15% damage | **3pc:** DEADEYE FORM, +25% damage, +20% homing *(weapon set — bonuses from computeGear)*

DEADEYE: 2pc +15% damage | 3pc DEADEYE FORM (+25% damage, +20% homing)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	De
◎	**DEADEYE SHOT**	14	70	30 frames	90 frames	aim	—	Pe sh dm Ao ne en
🎯	**HUNTER'S MARK**	8	150	30 frames	100 frames	player	—	Ma pr Ao yo ma

dragonlance

2pc: +1 pierce | **3pc:** DRAGON FORM, +2 pierce, +25% damage
(weapon set — bonuses from computeGear)

DRAGONLANCE: 2pc +1 pierce | 3pc DRAGON FORM (pierce +2, +25% damage)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	De
🔥	**DRAGON BREATH**	12	110	30 frames	90 frames	aim	burn	Dr fi dm Ao ne en
🐉	**TAIL SWEEP**	10	130	30 frames	100 frames	player	knockback	Ta 10 ar Kn ba

FROSTBOUND

2pc: 2pc: slows last +1s | **3pc:** 3pc: ICE FORM

FROSTBOUND: 2pc +60 frost | 3pc ICE FORM (frost always active, +1 max HP, enemies shatter)

Form passive: SIX-SET: Frostbound SHATTER (locked) — FORMULA: if iceForm AND enemy slowed (slowT > 0, pre-effect) at hit time, dd = dd x 1.50. Multiplicative on top of star/boss scaling. No stack (single 1.5x).

Ice Shard in full Frostbound form: 10 base / 15 effective — SHATTER (×1.5) always applies in full Frostbound form (no mechanic change; documented effective value).

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	Descri
❄	**ICE SHARD**	10	80	30 frames	60 frames	aim	slow	Pierci ice sh 10 dmg slow A neares enemy.
✦		8	130	30 frames		player	freeze	Freeze nova:

	FROST NOVA				120 frames			+ free AoE ar you.

MOONFANG

2pc: 2pc: +15% banana speed | **3pc:** 3pc: WEREWOLF FORM

MOONFANG: 2pc +15% bullet speed | 3pc WEREWOLF (+30% armor, +25% bullet speed, +2 pierce)

Form passive: MOONFANG BLOOD BARRIER (locked) — FORMULA: lifesteal heal +0.03 HP per enemy hit by a set skill while werewolf. Any heal that would EXCEED maxHP overflows into bloodBarrier (temporary HP). Cap = 2 x maxHP (scales with max HP) — total effective HP = 3 x maxHP (tripled, "feel like a real werewolf"). Damage order: Shield bubble (blocks whole hit first) -> bloodBarrier absorbs final damage (after all reductions) -> HP. Barrier persists until consumed (no decay). Reset each wave (does NOT carry over) + on new game / NG+ advance / replay. Drawn as a dark-red ring around Ned + 'B+N' segment in the HUD hearts row.

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	Des
☾	**MOON CLAW**	8	90	30 frames	60 frames	aim	bleed	Sil sla dmg nea ene cas Ble (Do
◉	**LUNAR HOWL**	12	140	30 frames	240 frames	player	knockback	Moo sho 12 aro Kno bac

starweaver

2pc: +20% homing | **3pc:** STAR FORM, +30% homing, +20% bullet speed *(weapon set — bonuses from computeGear)*

STARWEAVER: 2pc +20% homing | 3pc STAR FORM (homing max, +20% bullet speed)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	De
✦	**STARFALL**	10	100	30 frames	75 frames	aim	—	Fa st dm ne en Ho sh
✧	**NOVA BURST**	8	130	30 frames	90 frames	player	—	St 8 ar Bl ne fo

stormcaller

2pc: +15% AoE | **3pc:** STORM FORM, +25% AoE, +15% damage
(weapon set — bonuses from computeGear)

STORMCALLER: 2pc +15% AoE | 3pc STORM FORM (AoE +25%, chain lightning)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect
ϟ	**LIGHTNING STRIKE**	11	90	30 frames	70 frames	aim	—
☁	**THUNDERCLAP**	9	140	30 frames	110 frames	player	knockbac

SUNFIRE

2pc: 2pc: burns last +1s | **3pc:** 3pc: SUNFIRE FORM

SUNFIRE: 2pc +60 pyro | 3pc SUNFIRE FORM (fire always active, +20% damage, burn spreads)

Form passive: SIX-SET: Sunfire BURN STACKS (locked) — FORMULA: stacks = min(5, stacks+1) per burn application; instant bonus = 2 x stacks flat dmg (max bonus 10). burnT refreshed to 120 (existing 2s death timer). Bonus is FLAT (no star/gear scaling).

Right-click formula (METEORS): FORMULA (Meteors): 4 meteors, delays 0/10/20/30 frames, random points within 140px of the click (x0.8), sub-radius 55px each. Per strike: impact = 5 x skillStarMul; boss: ceil(x1.5 x (1 + 0.5*mods.bane + gear.bane)). Each hit: burnT=120 + Sunfire BURN STACKS (stacks cap 5, +2 flat per stack, max +10).

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	Des
☀	**SOLAR FLARE**	10	100	30 frames	75 frames	aim	burn	Sol met dmg AoE nea ene
☄	**METEORS**	5	140	30 frames	110 frames	aim	meteors	Met rai whe cli met dmg eac AoE

TEMPEST

2pc: 2pc: +15% banana speed | **3pc:** 3pc: TEMPEST FORM

TEMPEST: 2pc +15% bullet speed | 3pc TEMPEST FORM (+20% bullet speed, +15% AoE, chain lightning)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect
ϟ	**CHAIN LIGHTNING**	7	100	30 frames	70 frames	aim	chain

☁	**THUNDER SURGE**	9	140	30 frames	110 frames	player	knockback

thunderbolt

2pc: +15% bullet speed | **3pc:** THUNDER FORM, +30% bullet speed, +15% fire rate *(weapon set — bonuses from computeGear)*

THUNDERBOLT: 2pc +15% bullet speed | 3pc THUNDER FORM (+30% bullet speed, +15% fire rate)

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	Desc
⚡	**VOLT SPIKE**	9	85	30 frames	50 frames	aim	—	Fast spik dmg near enem Quic
◉	**STATIC FIELD**	7	120	30 frames	80 frames	player	slow	Stat fiel dmg AoE you.

VENOM

2pc: 2pc: +1 pierce | **3pc:** 3pc: SWARM FORM

VENOM: 2pc +1 pierce | 3pc SWARM FORM (swarm attacks, +25% move speed, poison spreads)

Form passive: SIX-SET: Venom POISON SPREADS (locked) — proximity spread while alive, ONE-TIME ONLY. FORMULA: poison = 1 dmg/s x 3s = 3 total per application. Spread radius 80px. Spread can ONLY poison an enemy that has NEVER been poisoned (everPoisoned === false). Once an enemy has been poisoned by ANY source (blob, pool, Stinger, any

poison skill), spread can NEVER re-poison it. Only a direct ATTACK (blob hit, pool, Stinger, any poison skill) can re-poison it. Spread chains (A->B->C): a spread-poisoned enemy can spread to a never-poisoned enemy, but each enemy is spread-poisoned at most once. Dead enemies don't spread (poison dies with the host). No stack (still 1 dmg/s, never 2).

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	D
†	**POISON BLOB**	6	80	30 frames	60 frames	aim	poisonpool	V 6 a e l p (E t g p
✚	**SWARM STRIKE**	9	90	30 frames	90 frames	aim	swarm	S d n e s h n

VOID

2pc: 2pc: +1 max HP | **3pc:** 3pc: VOID FORM

VOID: 2pc +1 max HP | 3pc VOID FORM (+15% damage, +10% armor, heal on kill)

Form passive: SIX-SET: Void VOID CONSUMPTION (heal-on-kill, locked — REPLACES the aura, no dash dependency). FORMULA: heal = +0.5 HP per kill within 150px of the player (distance from death position to player). Cap maxHP. No scaling. No cooldown. Bosses included (adds spawn in boss fights, cap 2).

Icon	Skill	Damage	Radius	Telegraph	Cooldown	Target	Effect	Descr
◌	**VOID BOLT**	10	90	30 frames	70 frames	aim	drain	Void 10 dm neare enemy Drain

								of da dealt
◎	**SHADOW VEIL**	0	0	30 frames	600 frames	player	veil	Shado no da Move slowe INVUL 5s. 1 coold

Transformation sprites — all 6 forms

Each form has a DISTINCT transformation sprite: the player character redrawn for that form. Tempest + Void were missing (they fell back to the base sprite) and were generated for this build from the base sprite with an elemental palette shift.

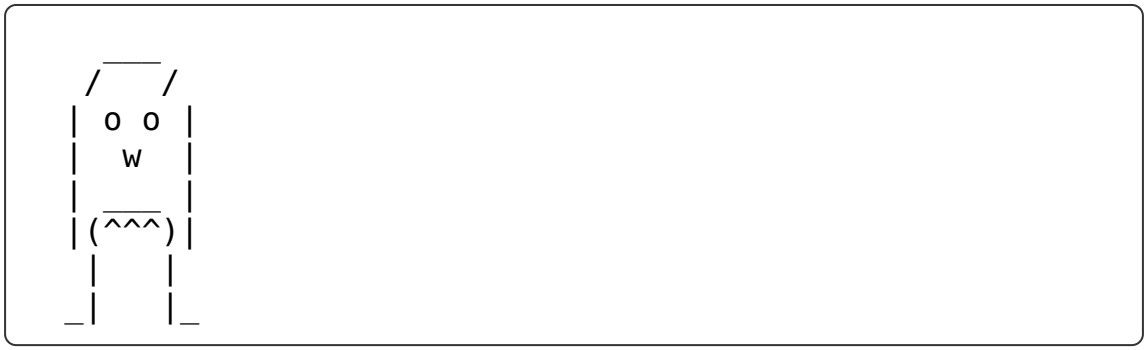
Moonfang (Werewolf)	Sunfire	Frostbound	Venom (Swarm)	Tempest	Void
Moonfang	Sunfire	Frostbound	Venom / Swarm	Tempest	Void

8. Bosses + Boss Skills

Each boss has 3 base attacks + 1 SIGNATURE (its defining move). Rotation: **P1 (NG+0) = [base0, base1, signature]** — P1 withholds the 3rd base attack. **P2/P3 (NG) = [base0, base1, base2, signature]** — NG restores the withheld base. The signature is a ROTATION member from fight start (100% HP), never withheld, NOT a one-time 50% HP trigger. Boss HP, projectile damage, and contact damage scale by WAVE with a HYBRID RAMP after wave 12 (see Formulas). Bosses have an element, a weakness (2x damage), and a resistance (0.5x damage). HP is NEVER shown. Actual boss HP = bossHPForWave(wave) × 1.6^ngp × 0.75 × 1.25 × tough (see §12); the hp field in the boss definitions is legacy and is NOT used at runtime. Waves 30–31 bosses also use ONE-HIT KILL (see Telegraphs). **Boss ATTACKS: see §21** (theme-specific spec, IMPLEMENTED 2026-10-14). The game now runs the theme-specific attacks (all 84 in execBossAtk; verified via forced attack cycles, 0 frame errors).

BOSS DEEPENING (implemented): every boss has a UNIQUE element + weakness combo, ONE signature mechanic no other boss has, and a UNIQUE telegraph shape (11 new shapes: ellipse, pentagon, heptagon, octagon, parallelogram, trapezium, cross, arrow, crescent, quatrefoil, heart). **Boss ATTACKS: see §21** — each boss’s 4 attacks DO something only that boss does (the mechanic IS the theme). PIP/POP: duo kit in §21 (Pea Fan + Mirror, Bully Charge + Squabble, Swap, Betrayal).

THE BANANA KING — all peels bow



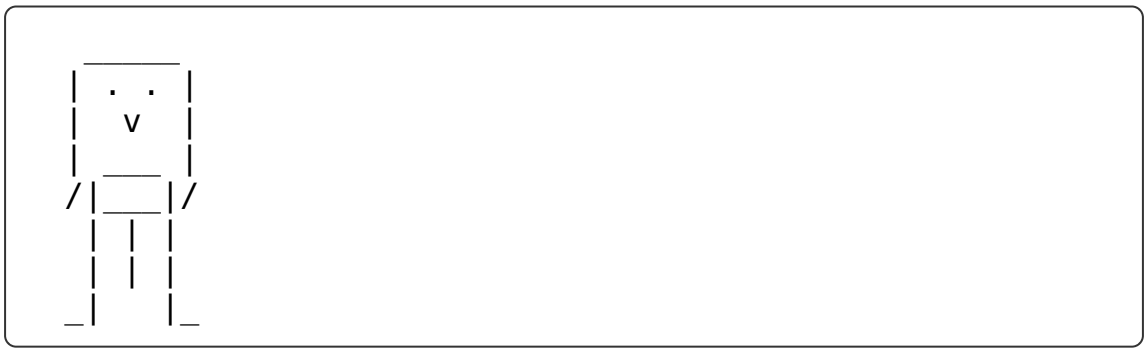
Animation (telegraph shape): quatrefoil

Colour: #ffd700 **Element:** fire **Weak:** physical **Resist:** void **Telegraph shape:** quatrefoil **HP:** wave-scaled (see §12) **Points:** 15000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: goldenpeel, worldsplitter, fangmoon, cinderblade, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

CAPTAIN BRINEBEARD — scourge of the peels



Animation (telegraph shape): triangle

Colour: #4d9fff **Element:** ice **Weak:** lightning **Resist:** fire **Telegraph shape:** triangle **HP:** wave-scaled (see §12) **Points:** 6500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: —, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE CONDUCTOR — every end has a score



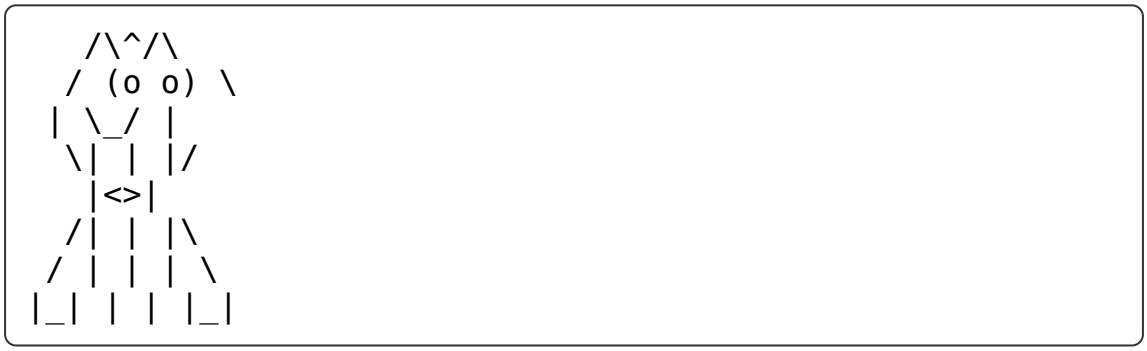
Animation (telegraph shape): hexagon

Colour: #d9a7ff **Element:** lightning **Weak:** ice **Resist:** fire **Telegraph shape:** hexagon **HP:** wave-scaled (see §12) **Points:** 7000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: baton, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE CRYSTAL WARDEN — every facet cuts



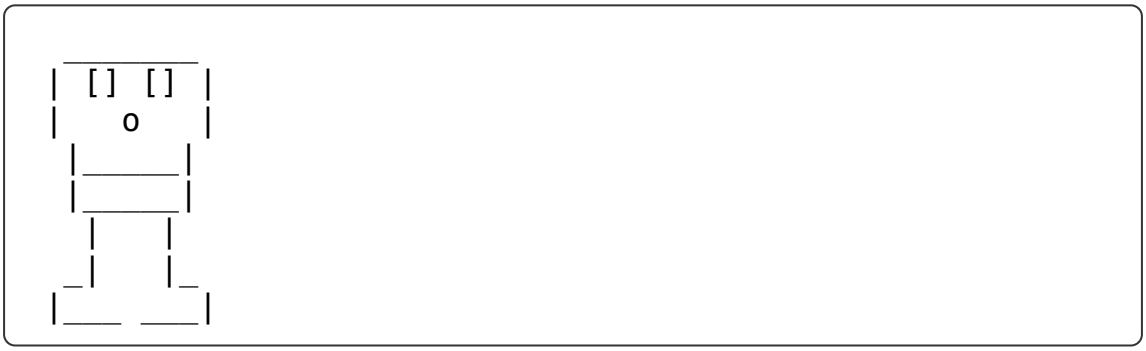
Animation (telegraph shape): beam

Colour: #ccf5ff **Element:** ice **Weak:** fire **Resist:** ice **Telegraph shape:** beam **HP:** wave-scaled (see §12) **Points:** 3300

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: prismcore, crystalshard, wardenlens, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE DREADNOUGHT — iron does not yield



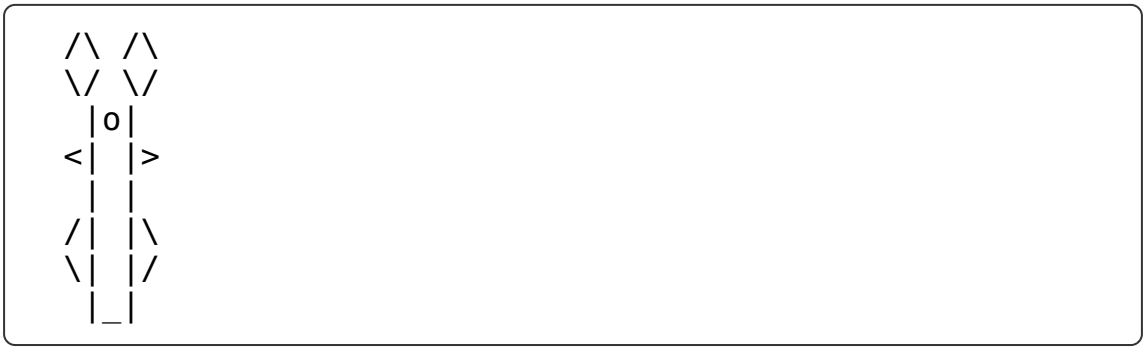
Animation (telegraph shape): arrow

Colour: #8899aa **Element:** fire **Weak:** ice **Resist:** physical **Telegraph shape:** arrow **HP:** wave-scaled (see §12) **Points:** 12500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: —, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

FROSTBITE — the silent glacier



Animation (telegraph shape): equilateral

Colour: #9df3ff **Element:** ice **Weak:** fire **Resist:** lightning **Telegraph shape:** equilateral **HP:** wave-scaled (see §12) **Points:** 4000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: glacialshard, frostblade, glacialaegis, wintersheart, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

GARGANTUA — he is the mountain



Animation (telegraph shape): square

Colour: #a08060 **Element:** physical **Weak:** void **Resist:** fire **Telegraph shape:** square **HP:** wave-scaled (see §12) **Points:** 11000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: —, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

HONEYBEAR, THE SNACK KING — sticky and sweet



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Animation (telegraph shape): rect

Colour: #ffcc55 **Element:** physical **Weak:** ice **Resist:** void **Telegraph shape:** rect **HP:** wave-scaled (see §12) **Points:** 3500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: honeydipper, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE JESTER — giggle doom

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Animation (telegraph shape): right

Colour: #ff6fd8 **Element:** void **Weak:** physical **Resist:** fire **Telegraph shape:** right **HP:** wave-scaled (see §12) **Points:** 5000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: squeakhammer, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE ABYSSAL LEVIATHAN — the deep does not forgive

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Animation (telegraph shape): diamond

Colour: #4d9fff **Element:** void **Weak:** lightning **Resist:** ice **Telegraph shape:** diamond **HP:** wave-scaled (see §12) **Points:** 3500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: leviathanscale, abysspearl, tidecaller, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE LIBRARIAN — silence is mandatory

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Animation (telegraph shape): parallelogram

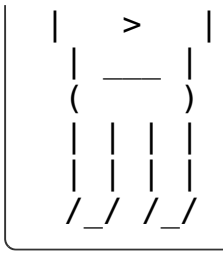
Colour: #c9b18a **Element:** ice **Weak:** void **Resist:** lightning **Telegraph shape:** parallelogram **HP:** wave-scaled (see §12) **Points:** 9500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: —, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

MOONHOWL, THE WOLF — he hunts the moon

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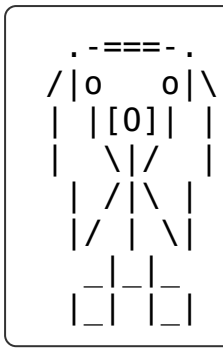
Animation (telegraph shape): crescent

Colour: #8f9bff **Element:** physical **Weak:** lightning **Resist:** ice **Telegraph shape:** crescent **HP:** wave-scaled (see §12) **Points:** 5500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: fangmoon, peltmight, mooncollar, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE CLOCKWORK ORACLE — time is a gear



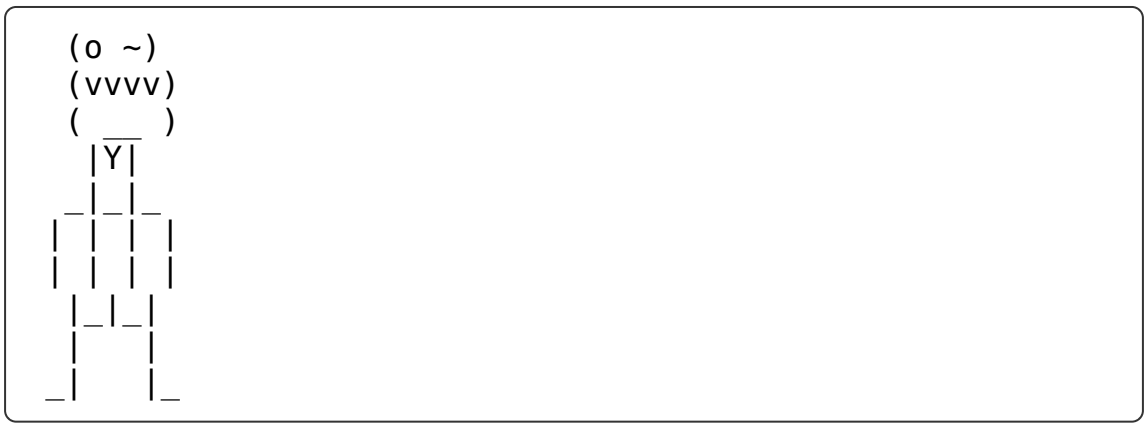
Animation (telegraph shape): heptagon

Colour: #ffd700 **Element:** lightning **Weak:** ice **Resist:** void **Telegraph shape:** heptagon **HP:** wave-scaled (see §12) **Points:** 3000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: oraclegear, timepiece, brasskey, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

PIP, THE BETRAYER — he seemed so nice



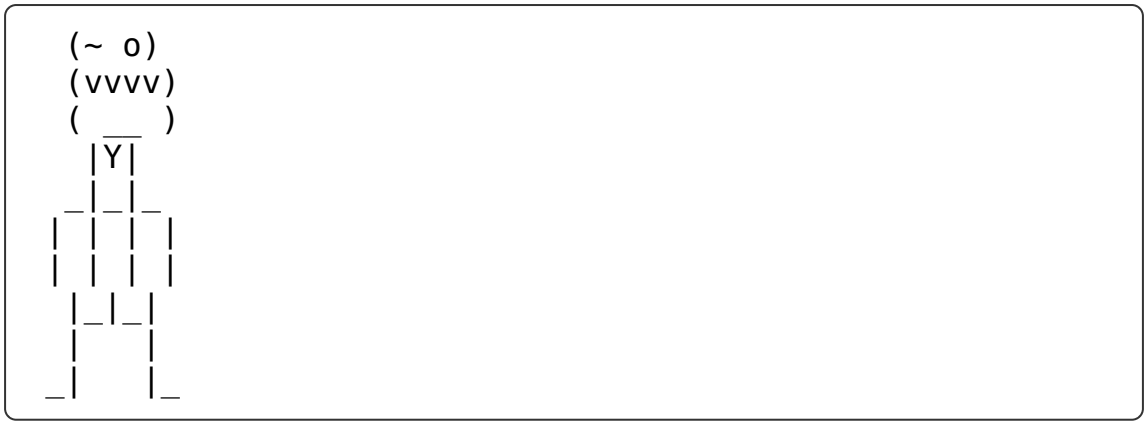
Animation (telegraph shape): cross

Colour: #7dff4d **Element:** fire **Weak:** void **Resist:** ice
Telegraph shape: cross **HP:** wave-scaled (see §12)
Points: 20000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: worldsplitter, stormwand, aegisdawn, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

POP, THE BETRAYER — she seemed so nice



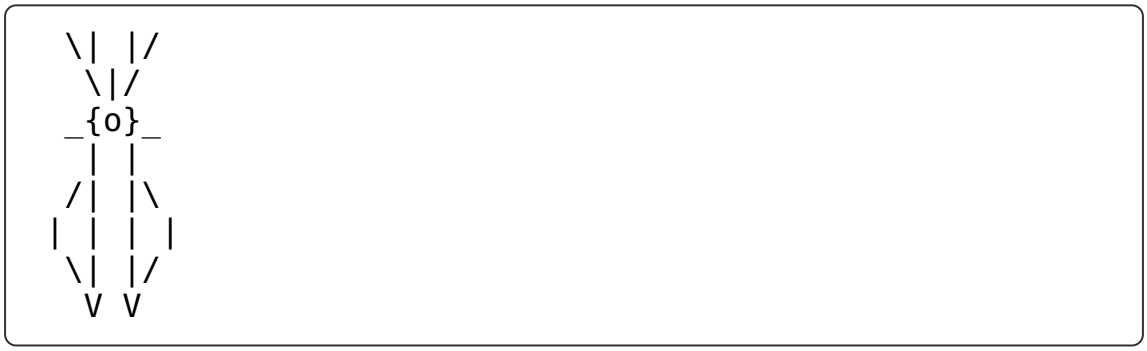
Animation (telegraph shape): ellipse

Colour: #ff7dff **Element:** void **Weak:** ice **Resist:** fire
Telegraph shape: ellipse **HP:** wave-scaled (see §12)
Points: 20000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: voiddagger, heartmountain, starwand, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

PYRE QUEEN — the burning throne



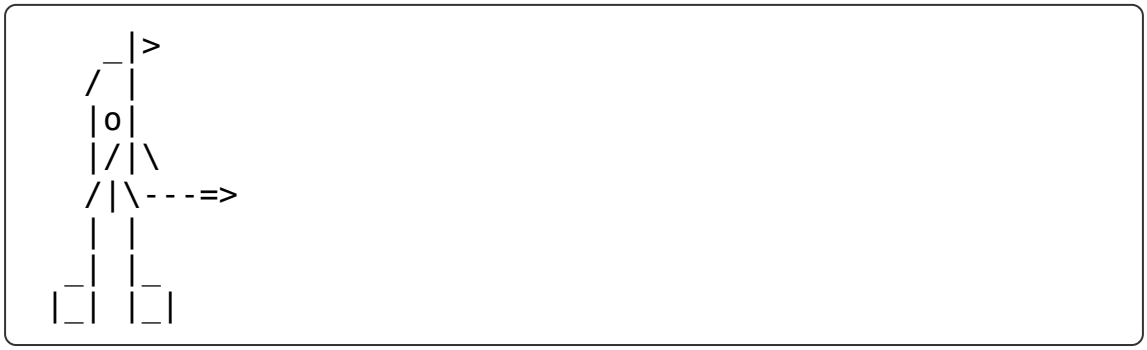
Animation (telegraph shape): heart

Colour: #ff7a2a **Element:** fire **Weak:** lightning **Resist:** ice **Telegraph shape:** heart **HP:** wave-scaled (see §12) **Points:** 2500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: cindercrown, cinderblade, ashenaegis, emberpendant, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

RONIN, THE FALLEN BLADE — swift sword saint



Animation (telegraph shape): iso

Colour: #cfe8ff **Element:** physical **Weak:** fire **Resist:** void **Telegraph shape:** iso **HP:** wave-scaled (see §12) **Points:** 1500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: fallenblade, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

STARWEAVER — she sews the night



Animation (telegraph shape): star

Colour: #d9a7ff **Element:** void **Weak:** lightning **Resist:** fire **Telegraph shape:** star **HP:** wave-scaled (see §12) **Points:** 14000

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: starwand, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE SUGAR WITCH — sweetness is a weapon



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Animation (telegraph shape): circle

Colour: #ff6fd8 **Element:** fire **Weak:** ice **Resist:** physical **Telegraph shape:** circle **HP:** wave-scaled (see §12) **Points:** 2800

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: candycane, gumball, sugarwand, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

THE THORN REGENT — the brambles remember

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Animation (telegraph shape): pentagon

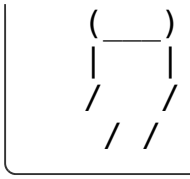
Colour: #37ff8b **Element:** physical **Weak:** fire **Resist:** ice **Telegraph shape:** pentagon **HP:** wave-scaled (see §12) **Points:** 3200

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: thorncrown, brambleblade, seedpouch, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

QUEEN VESPA — the stinging throne

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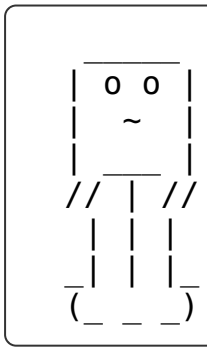
Animation (telegraph shape): octagon

Colour: #ffe14d **Element:** lightning **Weak:** physical **Resist:** void **Telegraph shape:** octagon **HP:** wave-scaled (see §12) **Points:** 7500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: stinger, venomfang, carapace, swarmheart, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

BARON VON WOBBLE — a most unstable noble



Animation (telegraph shape): trapezium

Colour: #ff9df2 **Element:** void **Weak:** fire **Resist:** lightning **Telegraph shape:** trapezium **HP:** wave-scaled (see §12) **Points:** 8500

Attacks: see §21 (theme-specific spec, IMPLEMENTED 2026-10-14).

Boss loot: count , high , lucky , signature: wobbleshield, sets: all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14)

9. Telegraphs

A telegraph is the warning BEFORE damage lands. Every boss attack, every elite (mini-boss) attack, and every set skill telegraphs. The telegraph

shape matches the projectile shape (circle / triangle / iso / equilateral / right / square / rect / hexagon / diamond / star) so the player reads it at a glance. Every boss has its own telegraph shape (see Bosses). **Boss-attack telegraphs follow the §21 theme-specific spec (IMPLEMENTED 2026-10-14)** — the game now uses the theme-specific telegraphs.

Telegraph	Timing	Shape	What happens
Boss windup	180 frames (3s)	Boss glows + shout (attack name, e.g. REWIND!)	Boss announces its next attack; the attack fires after the windup. Shout names follow the §21 spec (ATK_TELL, 87 keys, IMPLEMENTED 2026-10-14).
Meteor	210 frames	Boss shape (triangle / star / circle) at the player	A meteor falls on the telegraphed spot.
Cracks	Until eruption	Boss shape (crack lines) on the ground	Crack zones erupt; PHASE BOOTS walk through cracks.
Burn zones	Until burn ends	Boss shape, fire colour	Standing in a burn zone burns you. FIREPROOF = immune.
Slow zones	Until slow ends	Boss shape, ice colour	Standing in a slow zone slows you. STICKY FEET = immune + 10% speed in them.
Set-skill telegraph	30 frames (0.5s)	Circle AoE at aim point (or around player for nova)	Set skill lands after 0.5s; damage = skill dmg + radius.
Shush (Librarian A1)	180 frames (3s) telegraph at player	Boss shape (parallelogram) r=90 at player	If still inside on resolve: skills lock 3s (drop-skill effects off, set skills blocked). Plain peas still fire. HUD shows SILENCED Ns. (Moves the old SILENCE signature here; new signature is OVERDUE, see §21.)

Freeze (Time Stop / Freeze Nova)	180 frames (3s) telegraph at player (Time Stop); elite Freeze Nova freezes on hit	Heptagon freeze zone r=100 (Time Stop)	Inside on resolve = **frozen 1.5s (90f)** (cannot move/attack) while 5 gears fly. Cyan overlay on the player. Freeze Nova (Frost Guardian elite): 1 dmg + freeze 1s on hit.
Crush (Crushing Depths, Leviathan signature — KEPT, kyle’s pick)	180 frames (3s) crush telegraph at player	Diamond crush zone r=160	Crushes the player — **big damage** (atkMul × base, single hit). See §21.
Split (Ice Split, Crystal Warden A3)	180 frames telegraph at the boss	Beam split	On resolve: Warden splits into 2 (each 50% HP) for 5s — both fire fans, then recombine (HP = sum). (Replaces the old 1 → 3 shards split with a true split.)
Prison (Ice Prison, Crystal Warden signature)	180 frames (3s) telegraph at player	Beam cage r=100 at player	Crystal cage forms 3s (can move inside, cannot leave); then shatters — damage if still inside + 8 shards fly out. (Keeps ICE PRISON, upgraded: cage shatters into shards.)
Warn (elite / mini-boss telegraph)	60 frames (1s) wind-up before EACH elite attack (A and B)	Circle r=70, terrain colour, at the player	Boss-style telegraph for mini-bosses. Resolves to nothing — the attack fires. Regular trash (chaser, shooter, tank, spitter) has NO telegraphs; the

			projectile is the warning.
Slow (dur)	Slow zones carry a duration (default 360 = 6s)	Boss shape	Syrup Pool 240 (4s), Thorn Maze 300 (5s), Ice Prison cage 180 (3s), Honey Pool 240 (4s). Slow = chilled (move ×0.7) for the duration.
NG telegraphs	1s shorter (flat) in NG — P2 and P3 both −1s on boss + signature telegraphs (NOT per-tier)	Same shapes	OHK warning stays the full 6s (never shortened). P1 unchanged.
ONE-HIT KILL (waves 30–31 only)	180-frame windup shout + 180-frame telegraph (6s total warning)	Big flashing red/white circle (r=140) at the player's position when the attack fires	Still inside when it resolves = instant death (bypasses mercy invuln). Dodge out to survive. Dawnshield / shield-block / second-wind save you from a SINGLE OHK (earned counters). Each boss uses it once. DOUBLE OHK (Wave 31, ACCEPTED): Pip & Pop can both fire OHK on the same tick — their telegraphs are coincident (same tick, center, circle), so dodging one dodges both. But a single earned counter does NOT save you from a double OHK: inside a double OHK at resolution = death even with

			one counter. **OHK HP (locked 2026-10-14): CLAMP HP AT ZERO.** OHK overkill info does NOT matter (kyle: “does ohk information even matter” — no). HP never goes below 0 (display + metrics show 0, not negative — the earlier −1808 / −4993 negative HP metrics are fixed by clamping). The overkill info is discarded. IMPLEMENTED 2026-10-14 (if (lives < 0) lives = 0; in damagePlayer; verified via debug run).

Telegraph shape per boss

Boss	Shape
THE BANANA KING	quatrefoil
CAPTAIN BRINEBEARD	triangle
THE CONDUCTOR	hexagon
THE CRYSTAL WARDEN	beam
THE DREADNOUGHT	arrow
FROSTBITE	equilateral
GARGANTUA	square
HONEYBEAR, THE SNACK KING	rect
THE JESTER	right
THE ABYSSAL LEVIATHAN	diamond
THE LIBRARIAN	parallelogram
MOONHOWL, THE WOLF	crescent
THE CLOCKWORK ORACLE	heptagon

PIP, THE BETRAYER	cross
POP, THE BETRAYER	ellipse
PYRE QUEEN	heart
RONIN, THE FALLEN BLADE	iso
STARWEAVER	star
THE SUGAR WITCH	circle
THE THORN REGENT	pentagon
QUEEN VESPA	octagon
BARON VON WOBBLE	trapezium

circle

Telegraph shape: circle. AoE hitbox matches the shape (point-in-shape test).

triangle

Telegraph shape: triangle. AoE hitbox matches the shape (point-in-shape test).

iso

Telegraph shape: iso. AoE hitbox matches the shape (point-in-shape test).

equilateral

Telegraph shape: equilateral. AoE hitbox matches the shape (point-in-shape test).

right

Telegraph shape: right. AoE hitbox matches the shape (point-in-shape test).

square

Telegraph shape: square. AoE hitbox matches the shape (point-in-shape test).

rect

Telegraph shape: rect. AoE hitbox matches the shape (point-in-shape test).

hexagon

Telegraph shape: hexagon. AoE hitbox matches the shape (point-in-shape test).

diamond

Telegraph shape: diamond. AoE hitbox matches the shape (point-in-shape test).

star

Telegraph shape: star. AoE hitbox matches the shape (point-in-shape test).

ellipse

Telegraph shape: ellipse. AoE hitbox matches the shape (point-in-shape test).

pentagon

Telegraph shape: pentagon. AoE hitbox matches the shape (point-in-shape test).

heptagon

Telegraph shape: heptagon. AoE hitbox matches the shape (point-in-shape test).

octagon

Telegraph shape: octagon. AoE hitbox matches the shape (point-in-shape test).

parallelogram

Telegraph shape: parallelogram. AoE hitbox matches the shape (point-in-shape test).

trapezium

Telegraph shape: trapezium. AoE hitbox matches the shape (point-in-shape test).

cross

Telegraph shape: cross. AoE hitbox matches the shape (point-in-shape test).

arrow

Telegraph shape: arrow. AoE hitbox matches the shape (point-in-shape test).

crescent

Telegraph shape: crescent. AoE hitbox matches the shape (point-in-shape test).

quatrefoil

Telegraph shape: quatrefoil. AoE hitbox matches the shape (point-in-shape test).

heart

Telegraph shape: heart. AoE hitbox matches the shape (point-in-shape test).

beam

Telegraph shape: beam. AoE hitbox matches the shape (point-in-shape test).

10. Animations

Animations are the game’s movement + feedback. Each one is a named system in the code.

Animation	What it does
Muzzle flash	Brief flash at the muzzle in the projectile theme colour when firing.
Projectile flight	Peas fly toward the mouse; homing peas curve toward enemies; split peas divide on hit.
Fire trail (BLAZE)	Flickering glow/trail behind fire projectiles.
Explosion	Bomb / AoE burst: expanding ring + particles, radius = base x boomScale x (1 + 0.2*mods.aoe + gear.aoe).
Burn	Fire DoT: enemy burns (orange ticks), burn spreads to 1 nearby with PYRO.
Slow / Freeze	Ice slows enemies (blue tint, slowT). Deep freeze shatters gold (99 dmg).
Shatter	Frozen gold enemy shatters (ice tier 2 vs gold).
Knockback	Set skills / LUNAR HOWL / THUNDERCLAP / SHIELD BASH push enemies back.
Dash	

	Boss dash (RONIN dash, MOONHOWL pounce, DREADNOUGHT ram, POP bully) - boss lunges at a point.
Blink	Teleport (JESTER blink, WOBBLE wobble, PIP mirror) - boss vanishes and reappears.
Spin	Boss spin (FROSTBITE blizzard, CONDUCTOR spiral, JESTER confetti, BANANAKING peelstorm) - spiralling projectiles.
Summon	Boss summons adds (MOONHOWL mooncall: 2 sprinters; VESPA swarm: 3 sprinters; BANANAKING goldenrush: 2 gold; PIP/POP callboss: another boss, phase 2+).
Phase banner	Wave-31 phase change banner (PIP/POP phase 2/3) + shout.
Shout	Boss attack tell (ATK_TELL text above the boss) + Pip/Pop dialogue.
Charge	Player charge-up (OVERCHARGE: gun charges faster) - hold to charge a stronger shot.
Hit flash	Enemy flashes on hit; damage numbers float up.
Death	Enemy death: gems / hearts / skill drops pop out; boss death opens equipment choice.
Door / Level complete	Door opens after the boss dies; walk through to the next level.
Vendor	Vek the vendor: shop UI, buy skills / modifiers / mystery equipment.
Lore	Pip/Pop lore scenes (Waves 1, 5, 10, 15, 20, 25, 30) - one line/beat at a time, speaker orbs, map-colour staging.

11. Loot Tables + Percentages

Boss loot (Waves 1–30)

UNIFORM (locked): ALL bosses, same counts at a given NG level. **NG+0: 3 drops, Pick 1** (unpicked are LOST). **NG+1 / NG+2: 4 drops, Pick 1** (unpicked are LOST). **Pick 1 reverses keep-all (kyle 2026-10-14, §20.25).** Final boss (Pip & Pop) stays Pick 2. **Boosted slots:** 1 boosted (high) slot; 10% lucky chance adds a 2nd boosted slot.

NG level	Drops	Pick	Boosted slots

NG+0 (P1)	3	1 (unpicked LOST)	1 (+10% lucky 2nd)
NG+1 / NG+2 (P2/ P3)	4	1 (unpicked LOST)	1 (+10% lucky 2nd)

Boss rarity roll (rollBossRarity)

kyle: mythic = 8%, set = 5% (was mythic 3%, set 1%). Rare + unique reduced proportionally to match (total still 100%).

kyle: mythic + set ONLY drop from Wave 16+ bosses in P1 (gate = ngp===0 && wave<16). Below Wave 16 in P1, a mythic/set roll re-rolls (no mythic/set from any drop - boss, variant, or reward). The guarantee also excludes mythic below Wave 16 in P1.

Rarity	Base	Boosted (high slot)
common	35%	32%
uncommon	26%	24%
rare	22%	27%
unique	4%	4%
mythic	8%	8%
set	5%	5%

Guarantee: at least one rare-or-better per boss. Re-roll is weighted: rare 85%, unique 13%, mythic 2% (mythic -> unique below Wave 16 in P1). No set in the guarantee.

Boss signature roll (trySignatureDrop)

5% chance per boss-loot roll to drop one of THAT boss's signature items (picked randomly from that boss's BOSS_LOOT signature list), at EVERY boss fight (Waves 1–30 via rollBossLoot, Wave 31 via rollOneItem) — not just Wave 31. The signature roll runs BEFORE the normal rarity roll. **P1 gate respected:** a set/mythic signature item is blocked when ngp===0 && wave<16 (same gate as the normal rarity roll) — no early set/mythic drops via signature in P1.

LOCKED RULE (IMPLEMENTED 2026-10-14): boss signature items come ONLY from the dedicated 5% roll. The normal unique/set rarity path EXCLUDES boss signature items (no double-dipping). Total signature-item chance ≈5%. See §20.14.

Wave 31 (final boss) loot

Wave 31 uses the per-boss BOSS_LOOT table (count / high / lucky / signature), NOT the tier rarity roll. The sets field is REMOVED (IMPLEMENTED, see §20.21) — all bosses can drop any set. See each boss's card in Section 8 for its Wave-31 loot row.

Mystery equipment (vendor, 2 slots)

Rarity-weighted, NEVER set items.

Rarity	High roll	Normal roll
rare	70%	12%
unique	18%	4%
mythic	9%	2%
set	3%	0%
common	0%	55%
uncommon	0%	27%

Vendor (rollShop)

Exactly 10 items: 8 slots + 2 mystery equipment. NO pagination. Each of the 8 slots rolls **50% drop skill** (cost 50, goes into a drop-skill slot, replaces oldest if full) / **50% permanent modifier** (from the Vendor Items table, cost = item cost). 2 mystery equipment: rarity-weighted, never set. JACKPOT: pay 120, get 150 gems.

Enemy drops

Enemies drop: **gems** (value 5), **gold nuggets** (value 25), **hearts** (heal, 5% per kill), **skill drops** (random drop skill, 12% per kill), **stale bread** (4% per kill). **Wave 1 (tutorial wave, kyle 2026-10-14, §20.25):** exactly ONE add drops a guaranteed random drop skill (first non-boss kill); after that, Wave 1 adds NEVER drop skills. Hearts/currency unchanged in Wave 1. Gem value x (1 + 0.25 x mods.greedy + gear.greedy). LUCKY CHARM: +5% skill drop chance, kills drop bonus gems. Enemy tiers: ELITE (hp x1.3, speed x1.05, points x1.5, dropBonus 0, skills 1), RARE (hp x1.5, speed x1.15, points x2, dropBonus 1, skills 2), UNIQUE (hp x2.0, speed x1.20, points x5, dropBonus 2, skills 2, signature 1).

STACKING RULE (all +percentage bonuses): +percentage bonuses stack with DIMINISHING RETURNS (harmonic). 2 stacks of +50% = +50% + 50%*(1/2) = +75%, NOT +100%. 3rd stack = +50%*(1/3). harm(n) = 1 + 1/2 + 1/3 + ... + 1/n. Bonus = base * harm(n). Vendor perm effects (max stacks) use harm(). Gear % stats (items + traits) use harmonic weighting (largest counts fully,

2nd half, 3rd third). Flat bonuses (+1 max HP, +1 pierce, durations) stay linear. Exponential mods (SWIFT HANDS 0.9^n, IRON SKIN 0.75^n, ADRENALINE 0.85^n) are already diminishing - left as-is. Why: vendor max perm effects can be REALLY strong if percentages stack linearly (10 stacks of +15% = +150%). Harmonic keeps them strong but sane.
MAXED feedback: When a vendor perm effect is maxed out, the shop card shows **[MAX] MAXED** (grayed out, no price). Clicking a maxed item shouts **"NAME ALREADY MAXED! (owned/max)"**. TIME WARP (was a no-op) now works: enemies spawn 20% slower per stack (harmonic).

12. Formulas

Every number the game runs on. If a formula changes, this section changes — that's how we see it.

Player

Stat	Formula
Max HP	min(30, mods.maxhp + gear.maxhp) — CAP 30. Base maxhp 15, lives 15.
Move speed	Base 0.68 (BASE_STATS.moveSpeed).
Fire cooldown	base = rapid2 ? 2 : rapid1 ? 3 : 4; mult = 0.9^mods.swift x (1 - min(0.5, gear.rate)); if lives < maxHP/2: mult *= 0.85^mods.adren; mult = max(0.5, mult); cool = max(2, round(base x mult))
Projectile speed	BAN_SPEED x theme.spd x (homing2 ? 1.15 : 1) x (1 + 0.15 x mods.bspd + gear.bspd) x twist.spd
Charge	Hold to charge; OVERCHARGE makes the gun charge faster.

Damage

Step	Formula
1. Base pea	dmg = 1 (pea base, size 8). THERMAL SHOCK (fire+ice) twist: dmg = 2.
2. Explosions	dmg += mods.demo (DEMOLITION: explosions +1 damage).
3. Boss bonus	vs boss: dmg *= (1 + 0.5 x mods.bane + gear.bane) (BOSS BANE: +50% dmg to bosses).

4. Elemental base	Non-physical: dmg *= 0.75 (ELEMENTAL_BASE_MUL).
5. Matchup	dmg *= ELEMENT_MATCHUPS[attacker][defender] (5x5 table below).
6. Weak / Resist	Weak: dmg *= 2. Resist: dmg *= 0.5. Never both (no-overlap rule).

Elemental matchup table (attacker → defender)

	Physical	Fire	Ice	Lightning	Void
physical	1	1	1	1	1.5
fire	1	1	1.5	0.5	0.5
ice	1	0.5	1	1.5	0.5
lightning	1	1.5	0.5	1	0.5
void	0.5	1.5	1.5	1.5	1

Status effects

Effect	Formula
Burn	fireE tier 1 = 120 frames, tier 2 = 60 frames. PYRO: burns last +1s, burn spreads to 1 nearby. THICK SKIN: burn ticks slower, -10% fire damage.
Slow / Freeze	slowT = 180 + 60 x mods.frostp + gear.frost. FROSTPROOF: slows last +1s, slowed foes 10% slower. QUICK FEET: immune to chill/slow. Ice tier 2 vs gold = shatter (99 dmg).
Poison	Poison = 1 dmg/s x 3s = 3 total per application. No stack (re-apply = refresh). FLAT (no star/ gear scaling). Full rules: Section 15.
Set-skill scaling	Set-skill damage = skill dmg x 1.25^avgStars. Vs boss: ceil(x1.5 x (1 + 0.5 x mods.bane + gear.bane)). See Section 7 for per-skill formulas.
Auto-weapon damage	autoWeaponDmg = R.base x T.dmgMul x variance — variance rolled once at drop (0.85–1.15), stored on the item. Wave-found does NOT affect damage. Then the usual hit math: x star, x bane vs bosses, x element base, x RPS, x weakness/ resistance, x2 if shocked.
Damage routing	ALL damage routes through barriers/shields first (no bypasses). Shield bubble (blocks a whole hit, consumed) → Blood Barrier (absorbs up to

	barrier amount) → HP. Overflow → HP. OHK bypasses bubble + barrier. See Section 16.
Explosion radius	R x boomScale x (1 + 0.2 x mods.aoe + gear.aoe). BIG BOOM: +20% blast radius.
Homing turn	(homing2 ? 0.12 : 0.06) x (1 + 0.5 x mods.scope + gear.homing) x homingTurn. SCOPE: +50% homing turn.
Pierce	pierce2 ? 999 : pierce1 ? 2 + mods.pierce + gear.pierce + (werewolf ? 2 : 0) : 0, + twist.pierce.

Enemies

Stat	Formula
Zombie HP	ceil(T.hp x 1.3^ngp x (1 + 0.12 x max(0, wave-1))) — ngp = New Game+ cycle.
Enemy cap	24 + min(16, wave) concurrent zombies (Wave 1 = 25, Wave 16+ = 40).
Boss spawn	At least **60 seconds (1 min)** + proximity (IMPLEMENTED 2026-10-14: 45s → 60s). The boss spawns at MAX(60s, player arrival at the boss spawn point) — NOT before 1 min, and NOT before the player walks to the spawn point. **NOTE (deferred, kyle: “for check later” 2026-10-14 ~19:12):** MAX is the implemented default; the max-vs-min interpretation (MAX = boss waits for both 1 min AND player arrival vs MIN = boss spawns at whichever comes first) is DEFERRED for later check. Proximity: boss spawns when the player is < 480 px from bossPt; reveal 30s → 45s (2700f); gate 3600f. Reason (kyle): if maps get bigger, it may take MORE than 1 min for the player to walk to the spawn point — the boss must wait until the player arrives. Adds cap 2 during boss fights; variant chance x0.25 during boss fights. After boss death: no new adds, immediate equipment selection, active adds remain, door available.
Progression	Defeating the boss is required to progress. Door opens after boss death.

Bosses

Stat	Formula
	bossHPForWave(w) = (20 + (min(max(w,1),30)-1) x 5 + (c > 12 ? (c-12) x 5 : 0)) x 3, c =

Boss HP (wave)	min(max(w,1),30) — HYBRID RAMP x3 (kyle 2026-10-14, §20.25): +5/wave to wave 12, then +10/wave, all x3. Wave 1 = 60, Wave 12 = 225, Wave 15 = 315, Wave 30 = 765 (cap, before 0.9375 x tough).
Boss HP (actual)	round(bossHPForWave(wave) x 1.6^ngp x 0.75 x 1.25 x (boss toughness || 1))
Boss projectile dmg	1 + (c-1) x 0.05 + (c > 12 ? (c-12) x 0.05 : 0), c = min(max(wave,1),30) — HYBRID RAMP. Wave 1 = 1.0, Wave 12 = 1.55, Wave 30 = 3.4.
Boss contact dmg	1.5 + (c-1) x 0.1 + (c > 12 ? (c-12) x 0.1 : 0) — HYBRID RAMP. Wave 1 = 1.5, Wave 12 = 2.6, Wave 30 = 6.3.
Boss phases	Waves 1–14: one phase. Waves 15–30: phase 2 at 50% HP. Wave 31: three discrete phases (bossPhase).

Currency / Drops

Stat	Formula
Gem value	round(5 x (1 + 0.25 x mods.greedy + gear.greedy)). Gold nugget: round(25 x (1 + 0.25 x mods.greedy + gear.greedy)). Currency is UNCAPPED until collected.
Greedy trait	+10% gem value. Lucky trait: +3% drop chance.

NG difficulty curve (implemented)

IMPLEMENTED (Slice B + C). The table below is what the game runs. The old flat +ngp damage taken is REMOVED. NG also makes foes faster and warnings shorter (see the last three rows).

NG scaling	Variant 1 formula (ngp = NG cycle)
Gear / star damage	x1.5^ngp — each star +50% gear damage (see Section 13). Drop skills +25% per star (1.25^avgStars).
Boss HP	x1.6^ngp — slightly faster than star damage, so boss fights lengthen gently each tier.
Non-boss enemy HP	x1.3^ngp — slower than star damage, so trash stays killable.
Enemy damage	x1.15^ngp — smooth pressure instead of a flat jump.

Flat +ngp damage taken	REMOVED — the old flat +ngp damage taken per hit goes away.
Enemy speed (NG)	x(1 + ngp x 0.05) — +5% per NG tier.
Projectile speed (NG)	x(1 + ngp x 0.05) — +5% per NG tier.
Telegraphs (NG)	1s shorter (flat) in NG — P2 and P3 both −1s on boss + signature telegraphs (NOT per-tier). OHK warning stays 6s.
Cap (P3 / NG+2 max)	THREE PLAYTHROUGHS total: P1 = NG+0, P2 = NG+1, P3 = NG+2 (FINISH). Max scaling at ngp=2: star x2.25, boss HP x2.56, trash HP x1.69, enemy damage x1.3225. Stars MAX 2. There is no NG+3.

Design goal: FLOW STATE. Power fantasy (stars) + scaling difficulty = challenge matches skill. Never too easy (boredom), never too hard (anxiety).

Performance: cartridge-era optimization (Nintendo up to N64 discipline)

OPTIMIZE = cartridge-era efficiency (locked). The game runs at 60fps on limited power + storage, so it squeezes the most out of what it has. **No allocations in the hot loop:** pooled projectiles / trail points / trail arrays (reused, never re-allocated per frame; pools degrade gracefully — if empty, a fresh object is allocated). **In-place loops:** backward index loops instead of [...arr] copies, filter/map/slice replaced by in-place compaction in hot paths. **Cached lookups:** playerRadius() cached per frame; ZTYPES[type] cached per zombie. **Lookup tables:** 1024-entry sin/cos table for visual / weave / polygon-point math (never for shot-direction math). **Distance-squared** instead of sqrt where only a comparison is needed. **Early-outs:** cheap bbox check before expensive inShape branches (all shapes fit in 1.5r except line). **No string building** in hot paths. **NOT minified, comments NOT stripped** — the file stays readable (comments are the docs). File size is reported, never chased at the cost of readability.

13. New Game+ (NG) System

Three playthroughs total: **P1 = NG+0, P2 = NG+1, P3 = NG+2 (FINISH)**. There is no NG+3. Save version v18 persists ngp, tier, item stars, equipment, run stash, and the persistent stash. Older saves load with stars defaulting to 0 and the persistent stash unlocked iff ngp >= 1.

The prize: beating the last boss

On victory (beating the last boss), choose **REPLAY** or **ADVANCE**.

Victory choice	Rule
REPLAY (same difficulty)	Keep items **AS-IS** (equipped + stash, stars unchanged). Lose mods + currency. NO star update, **no prize screen**. Play the same tier again.
ADVANCE (next tier)	**Persistent-stash screen opens FIRST** (state 14, §20.26: deposit/withdraw run ↔ persistent stash; DONE goes to prize). Prize screen opens **before** NG starts (prize below). Lose unpicked items, reset mods/gems/slots. NG+0→+1, +1→+2. No ADVANCE past +2 (P3 is the finish).

The prize (ADVANCE only):

Prize step	Rule
Pick 3 items to keep	Pick any **3 items** to keep for NG from **all owned items** (equipped + stash). Everything else is **lost**. This is the real choice: best weapon, set pieces, or survival gear?
Pick 1 item to star	Pick **1 of those 3** to become **starred** at the new tier’s star count (stars = min(2, ngp+1)). The other 2 kept items keep whatever stars they already have. NOT all equipped items auto-starred — only the 1 you pick.
Kept items auto-equip	The 3 kept items **auto-equip** at the start of the next playthrough (restores the pre-ADVANCE loadout — your choice, not a “best items” pick). Neck → neck; hands fill LEFT then RIGHT in kept order. Any kept item that cannot equip stays in the run stash.

On entering NG

What happens	Rule
Vendor mods	**Reset** to 0 (permanent abilities are lost).
Gems	**Reset** to 0 (fresh economy).
Drop-skill slots	**Reset** — re-pick per run.

Gear kept	Only the **3 picked items** (1 starred). Everything else is lost.
Run stash	**Wiped** — fresh empty stash each playthrough (per-playthrough, cap 6).
Persistent stash	**Untouched** — survives across NG (see below).

Persistent stash (unlocked after Playthrough 1)

The **persistent stash** is a second stash that **survives across NG**. It is **UNLOCKED only after finishing Playthrough 1** (victory at NG+0). During P1 there is no persistent stash — run stash only.

Rule	Detail
Cap	**18 SLOTS (6 × 3)** — LOCKED 2026-10-14, IMPLEMENTED 2026-10-14 (MAX_PERSISTENT_STASH = 18, 6×3 layout). Holds ALL 18 set items (6 sets × 3pc) + 3 equipped = 21 total. GOAL: collect all 18 set items, try each set, finish NG+2 with each new set. Drives REPLAYABILITY (replay the save file until all set items are collected).
Accessible	In **ALL replays** (P2, P3, …) once unlocked. Items in it survive every ADVANCE.
DEPOSIT	Move a run-stash item → persistent stash (before ADVANCE) to save it from the ADVANCE wipe. Only if the persistent stash has a free slot.
WITHDRAW	Move a persistent-stash item → run stash in any replay. Only if the run stash has a free slot.
Equipment screen	DEPOSIT / WITHDRAW modes in the equipment screen (ESC). Deposit and withdraw never touch equipped items.

Stars (★)

Rule	Detail
Star count = NG tier	P1 (NG+0) = 0 stars. P2 (NG+1) = 1 star. P3 (NG+2) = 2 stars. Stars MAX 2. There is no NG+3.
Star value	Gear **+50% damage per star** (1.5^stars). Drop skills **+25% damage per star** (1.25^avgStars). Both multiplicative.

ALL NG items are starred	Every drop, boss loot, and vendor item in NG comes starred at the current tier’s star count. Rarity and stars are separate — a starred common exists.

Sets + loot gate in NG

NO smart drops. Set completion comes from kept partial sets + the open loot gate + a full run of rolls to find the missing piece. No weighting toward sets you own.

Loot gate LIFTED in NG: level 1 in NG can drop **uniques, mythics, and set items**. In P1 (NG+0), mythic/set are gated to Wave 16+ (see Section 11).

NG tutorial (victory screen, after the first playthrough — not front-loaded)

Verbatim tutorial text shown on the victory screen:

ADVANCE: keep 3 items (rest LOST), star 1. Kept 3 AUTO-EQUIP. Skill slots reset. Mods + gems reset.
REPLAY: keep ALL items as-is (stars unchanged). Same difficulty. Mods + gems reset.
Stars +50% gear / +25% skill per star. P2 drops ★, P3 drops ★★. NG foes faster, warnings shorter.

Wave 0 hint (in-game tutorial)

FARM ADDS FOR DROP SKILLS: in Wave 1, kill the little adds first — they drop skills. Farm adds → grab drop skills → THEN fight the boss. — shown as a Wave 0 tutorial step and on the HOW TO PLAY screen.

P3 (NG+2) = game FINISH

THREE PLAYTHROUGHS total. At P3 (NG+2), bosses have **ALL mechanics + ALL attacks** at **max scaling**: gear x2.25, boss HP x2.56, trash HP x1.69, enemy damage x1.3225. Beating the game at P3 is the finish; after that, REPLAY P3 (same difficulty) or stop. There is no NG+3. P3 victory shows the finish screen + replay; no prize.

14. Design Philosophy: Build vs Skill

Standing balance principle. It governs tuning (including the NG curve in Section 12) but is not itself a mechanic.

Good build deserves to snowball. Power fantasy is earned through build quality — the right gear, the right choices, the right loot. A player who builds well SHOULD snowball.

Crap build loses — unless the player is good at dodging. Skill is the fallback that compensates for a bad build. Every mechanic stays dodgeable (telegraphed, fair patterns) so a skilled player with a weak build can still win.

Two paths to winning: build well, or dodge well. **Build + dodge together = snowball.**

15. Poison Rules

Poison = 1 dmg/s × 3s = 3 total per application. No stack (re-apply = refresh to 180 frames). **FLAT** (no star / gear scaling on the DoT).

Source	Kind	Rule
AoE BLOB (Venom LEFT CLICK)	ATTACK	6 dmg on hit + leaves a POISON POOL. Can re-poison ever-poisoned enemies. Sets everPoisoned = true.
POISON POOL (left by the blob)	ATTACK	Radius 80px, duration 180 frames (3s). Enemies standing in the pool take **direct 1 dmg/s** while inside it AND get poisoned (poisonT=180, refreshed while standing in it). The poison DoT (1 dmg/s × 3s) **continues after the enemy leaves the pool**. Sets everPoisoned = true.
POISON SPREADS (Venom passive)	SPREAD (one-time)	Proximity 80px, while alive. Can ONLY poison an enemy that has NEVER been poisoned (everPoisoned === false). Once an enemy has been poisoned by ANY source, spread can NEVER re-poison it. Only a direct ATTACK can re-poison it.
Stinger gear / poison skills	ATTACK	Shots / skills poison (poisonT=180). Sets everPoisoned = true. Can re-poison.

Spread chains (A→B→C): a spread-poisoned enemy can spread to a never-poisoned enemy, but each enemy is spread-poisoned **at most once**. This kills the infinite refresh loop (spread was OP). **Dead enemies don’t spread** (poison dies with the host). Per-enemy everPoisoned flag.

16. Damage Routing + OHK

Damage routing (locked): ALL damage through barriers/ shields first

No damage bypasses them (including trail damage — the pre-existing trail bypass was fixed). Order:

Step	What happens
1. Shield bubble (auto-weapon Shield item)	Blocks a WHOLE hit, consumed. 1 shield = 1 charge; 2 shields = 2 charges, CD = better (shorter) shield's CD. CD by rarity: common/ uncommon 10s, rare 8s, unique 6s, mythic 5s, set 4s. Does NOT stack with BANANA SHIELD (separate bubble, own 20s CD).
2. Blood Barrier (Moonfang passive)	Absorbs the FINAL damage (after iron/armor + NG reductions), up to the barrier amount. Overflow → HP (not negated). Persists until consumed (no decay). Reset on new game / NG+ advance / replay, AND each wave (genLevel resets the Barrier — it does NOT carry over).
3. HP	Remaining damage hits HP.

Moonfang passive: LIFE STEAL + BLOOD BARRIER

Life steal: **+0.03 HP per enemy hit by a set skill while werewolf** (IMPLEMENTED 2026-10-14: lifesteal 0.05 → 0.03; kyle: Option 1 — lower life steal, no decay). No scaling on the rate, no cooldown. Any lifesteal heal that would EXCEED max HP overflows into the Blood Barrier. **Barrier cap = 2 × max HP** (scales with max HP, tank feeling kept), so **total effective HP = 3 × max HP**. **Barrier RESETS each level (wave) — does NOT carry over** (per-wave: built during a wave via life steal, reset at the next wave). **NO decay** (no in-combat, no out-of-combat — the per-wave reset already kills banking between waves). Drawn: dark-red ring around the player + B+N segment in the HUD hearts row.

OHK (One-Hit Kill, Waves 30–31)

Warning: 180-frame windup shout + 180-frame telegraph = **6s total**. Big flashing red/white circle (r=140) at the player's position when the attack fires. Still inside on resolve = instant death (bypasses mercy invuln). Each boss uses it once. **Double OHK (Wave 31):** Pip & Pop can both fire OHK on the same tick — telegraphs coincide, so dodging one dodges both; one earned counter does NOT save vs a double OHK.

OHK vs	Result
Shield bubble (auto-weapon Shield item)	**BYPASSED** — OHK kills through the bubble (not blocked).
Blood Barrier (Moonfang)	**BYPASSED** — OHK kills through the barrier (not absorbed). Barrier is temp HP, not a block.
Aegis of Dawn (dawnshield)	**SAVES vs one OHK** (purposeful, earned save — stays).
Shield block (BANANA SHIELD vendor item)	**SAVES vs one OHK** (purposeful, earned save — stays). Note: this is the VENDOR shield, NOT the auto-weapon Shield bubble.
Second Wind	**SAVES vs one OHK** (purposeful, earned save — revive at 3 HP).

All NON-OHK damage still routes through bubble → barrier → HP (overflow → HP, no bypasses).

17. New Bosses (IMPLEMENTED)

The 5 new bosses are IN THE GAME (BOSS_DEF + BOSS_ORDER; also listed in §8) IMPLEMENTED 2026-10-14 (rotation fix, see §20.2): the rotation now uses %bossOrderRun.length, so all 20 rotation bosses appear; the 5 new bosses sit at rotation indices 15–19 and now spawn. Rotation rule (same as implemented bosses): P1 (NG+0) = [base0, base1, signature]; P2/P3 (NG) = [base0, base1, base2, signature]. The signature is a ROTATION member from fight start (100% HP), never withheld, NOT a one-time 50% HP trigger. Boss HP formula: bossHPForWave(wave) × 1.6^ngp × 0.75 × 1.25 × tough. Each boss has a UNIQUE life total via its tough multiplier (tough was chosen at design as hp/30 with Oracle baseline 30; the hp field in the boss definitions is legacy and is NOT used at runtime — actual HP is wave-scaled, see formula). Boss attack damage: bossProjDmg (wave ramp) × atkMul. Boss contact damage: bossContactDmg (wave ramp). LEVIATHAN LOOT IS STANDARD (2026-10-14, kyle: "reduce to normal"): count 3 / high 1 / lucky 0.10 — the same as Oracle / Sugar Witch / Thorn Regent. NO exception for hardest boss — follow the loot table guidelines. CRYSTAL WARDEN is ICE: Oracle stays the only lightning boss.

THE CLOCKWORK ORACLE

time is a gear — Shape: **heptagon** — Element: **lightning** — Weak: **ice** — Resist: **void**

HP: **wave-scaled** (see formula below; the hp field in the boss definitions is legacy, NOT used) | Points: **3000** | Tough: **1.00** (unique life total) | Attack multiplier: **1.15** | Color: #ffd700

Formula: HP = bossHPForWave(wave) × 1.6^ngp × 0.75 × 1.25 × 1.00. Attack damage = bossProjDmg (wave ramp) × 1.15. Contact damage = bossContactDmg (wave ramp).



Attacks (theme-specific spec, IMPLEMENTED)

See §21 (theme-specific boss attacks, IMPLEMENTED 2026-10-14 — all 84 attacks in execBossAtk, verified via debug run).

Loot (BOSS_LOOT)

count 3, high 1, lucky 0.10 — signature: **oraclegear, timepiece, brasskey** — sets: **all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14; per-boss set restrictions removed)**. Signature roll: 5% chance per boss-loot roll to drop one of this boss's signature items. P1 gate applies (signature set/mythic blocked when ngp==0 && wave<16).

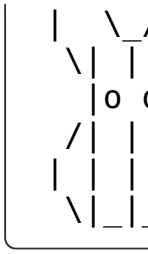
THE SUGAR WITCH

sweetness is a weapon — Shape: **circle** — Element: **fire** — Weak: **ice** — Resist: **physical**

HP: **wave-scaled** (see formula below; the hp field in the boss definitions is legacy, NOT used) | Points: **2800** | Tough: **0.93** (unique life total) | Attack multiplier: **1.10** | Color: #ff6fd8

Formula: HP = bossHPForWave(wave) × 1.6^ngp × 0.75 × 1.25 × 0.93. Attack damage = bossProjDmg (wave ramp) × 1.10. Contact damage = bossContactDmg (wave ramp).





Attacks (theme-specific spec, IMPLEMENTED)

See §21 (theme-specific boss attacks, IMPLEMENTED 2026-10-14 — all 84 attacks in execBossAtk, verified via debug run).

Loot (BOSS_LOOT)

count 3, high 1, lucky 0.10 — signature: **candycane, gumball, sugarwand** — sets: **all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14; per-boss set restrictions removed)**. Signature roll: 5% chance per boss-loot roll to drop one of this boss's signature items. P1 gate applies (signature set/mythic blocked when ngp==0 && wave<16).

THE THORN REGENT

the brambles remember — Shape: **pentagon** — Element: **physical** — Weak: **fire** — Resist: **ice**

HP: **wave-scaled** (see formula below; the hp field in the boss definitions is legacy, NOT used) | Points: **3200** | Tough: **1.20** (unique life total) | Attack multiplier: **1.15** | Color: #37ff8b

Formula: HP = bossHPForWave(wave) × 1.6^ngp × 0.75 × 1.25 × 1.20. Attack damage = bossProjDmg (wave ramp) × 1.15. Contact damage = bossContactDmg (wave ramp).



Attacks (theme-specific spec, IMPLEMENTED)

See §21 (theme-specific boss attacks, IMPLEMENTED 2026-10-14 — all 84 attacks in execBossAtk, verified via debug run).

Loot (BOSS_LOOT)

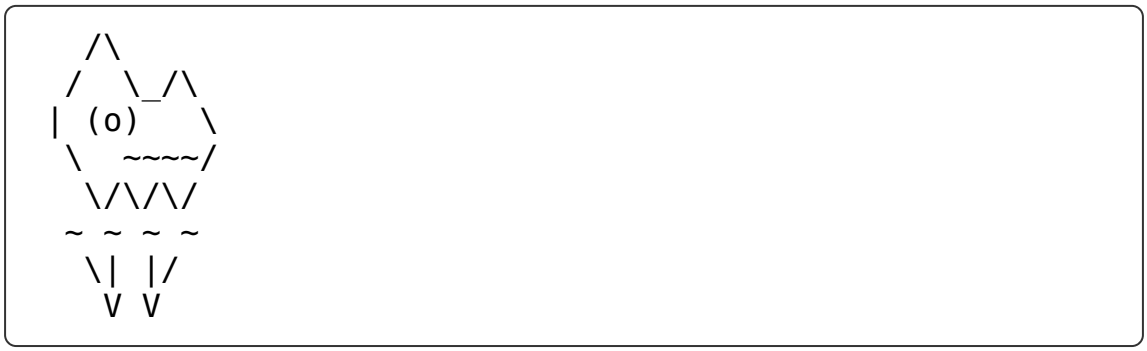
count 3, high 1, lucky 0.10 — signature: **thorncrown, brambleblade, seedpouch** — sets: **all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14; per-boss set restrictions removed)**. Signature roll: 5% chance per boss-loot roll to drop one of this boss's signature items. P1 gate applies (signature set/mythic blocked when ngp==0 && wave<16).

THE ABYSSAL LEVIATHAN

the deep does not forgive — Shape: **diamond** — Element: **void** — Weak: **lightning** — Resist: **ice**

HP: **wave-scaled** (see formula below; the hp field in the boss definitions is legacy, NOT used) | Points: **3500** | Tough: **1.33** (unique life total) | Attack multiplier: **1.20** | Color: #4d9fff

Formula: HP = bossHPForWave(wave) × 1.6^ngp × 0.75 × 1.25 × 1.33. Attack damage = bossProjDmg (wave ramp) × 1.20. Contact damage = bossContactDmg (wave ramp).



Attacks (theme-specific spec, IMPLEMENTED)

See §21 (theme-specific boss attacks, IMPLEMENTED 2026-10-14 — all 84 attacks in execBossAtk, verified via debug run).

Loot (BOSS_LOOT)

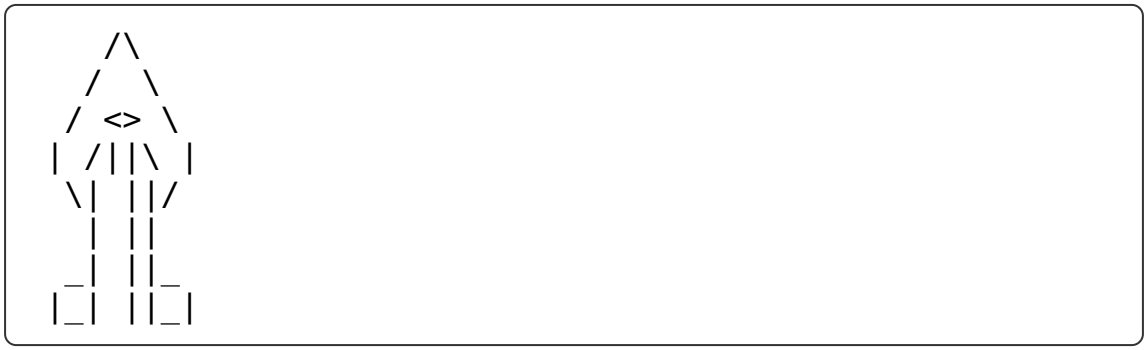
count 3, high 1, lucky 0.10 — signature: **leviathanscale, abysspearl, tidecaller** — sets: **all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14; per-boss set restrictions removed)**. Signature roll: 5% chance per boss-loot roll to drop one of this boss's signature items. P1 gate applies (signature set/mythic blocked when ngp==0 && wave<16). **Leviathan loot is STANDARD (2026-10-14): no exception for hardest boss.**

THE CRYSTAL WARDEN

every facet cuts — Shape: **beam** — Element: **ice** — Weak: **fire** — Resist: **ice**

HP: **wave-scaled** (see formula below; the hp field in the boss definitions is legacy, NOT used) | Points: **3300** | Tough: **1.13** (unique life total) | Attack multiplier: **1.20** | Color: #ccf5ff

Formula: HP = bossHPForWave(wave) × 1.6^ngp × 0.75 × 1.25 × 1.13. Attack damage = bossProjDmg (wave ramp) × 1.20. Contact damage = bossContactDmg (wave ramp).



Attacks (theme-specific spec, IMPLEMENTED)

See §21 (theme-specific boss attacks, IMPLEMENTED 2026-10-14 — all 84 attacks in execBossAtk, verified via debug run).

Loot (BOSS_LOOT)

count 3, high 1, lucky 0.10 — signature: **prismcore, crystalshard, wardenlens** — sets: **all 6 sets (any boss can drop any set — IMPLEMENTED 2026-10-14; per-boss set restrictions removed)**. Signature roll: 5% chance per boss-

loot roll to drop one of this boss's signature items. P1 gate applies (signature set/mythic blocked when ngp==0 && wave<16).

18. New Trash Mobs by Terrain (IMPLEMENTED)

These 35 mobs (5 per terrain × 7 terrains) are IN THE GAME (ZTYPES + spawn pools + terrain assignment). 5 archetypes per terrain (one chaser, one shooter, one tank, one spitter, one elite mini-boss). Mob HP formula: base HP × 1.3^ngp (non-boss HP scaling). Mob damage formula: listed dmg (flat; trash damage does not scale with NG). Contact damage = listed contact (flat; tanks/elites 2, others 1). Values below are wave-0, NG+0. TELEGRAPHS: elites ONLY (mini-boss rule) — boss-style telegraph (60f / 1s wind-up, circle r=70, terrain colour) before EACH attack (A and B). Regular trash (chaser, shooter, tank, spitter) has NO telegraphs — the projectile is the warning, consistent with existing trash. SPAWN GATES (buildLevelQueue, same pattern as the existing pool — tank W2, sprinter W3, ghost W4): chaser/shooter/spitter from Wave 1, tank from Wave 2 (same as base tank), elite from Wave 3 (LOCKED 2026-10-14; already satisfied — variantChance already gated elite at Wave 3). Tank ARMOR BLOCK: physical damage ×0.5 (elemental unaffected), with a shield/block flash on hit (hit feedback, not a telegraph). Armor block applies to pea/gear physical hits only (hitZombie); set-skill AoE, explode, and DoT ticks do NOT apply it. Chaser trail specials are flavor text (no trail zones).

Terrains (7)

Terrain	Element	Weak / Weak2	Resist / Resist2	Colour	Note
GREEN MEADOW	physical	fire / lightning	void / ice	#37ff8b	Grass, flowers, wind. Meadow mobs zigzag, hide in tall grass, and sing.
DESSERT DESERT	fire	ice / void	physical / fire	#ff8c1a	Candy, sugar, sticky. Dessert mobs are sweet, sticky, and burn on contact.

FROST CAVES	ice	fire / physical	lightning / ice	#9df3ff	Ice, snow, wind. Frost mobs slow, freeze, and shatter.
STORM PEAK	lightning	ice / physical	lightning / void	#ffe14d	Static storms crackle (3 storm cells shock you). Lightning mobs.
HONEY MARSH	void	lightning / fire	physical / void	#aaff00	Honey, mud, swamp gas. Marsh mobs stick, poison, and bog.
THORNWOOD	physical	fire / lightning	ice / void	#2a8f4d	Brambles, vines, roots. Thorn mobs root, lash, and bleed.
VOID CASTLE	void	lightning / physical	physical / void	#b48cff	Void, null, entropy. Void mobs phase, drain, and collapse.

GREEN MEADOW — 5 mobs

Grass, flowers, wind. Meadow mobs zigzag, hide in tall grass, and sing. Element: **physical**. Mob HP = base HP × 1.3^ngp. Mob damage = listed dmg (flat; trash damage does not scale with NG). SPEED MAPPING: the Speed column is the SPEC abstract speed; the game maps it to px/frame: chaser 1.2 → 0.35,

shooter 0.7 → 0.18, tank 0.45 → 0.08, spitter 0.6 → 0.14, elite 0.6 → 0.14 (before wave/NG scaling).

Name	HP	Element	Weak	Resist	Speed	Contact	Attack	Beha
Grass Runner	2	physical	fire	void	1.20	1	0	Runs the play a zi
Daisy Shooter	2	physical	lightning	ice	0.70	1	1	Keep dist fire peta bol
Boulderback	8	physical	fire	void	0.45	2	0	Slow arm with moss roc
Pollen Spitter	3	physical	fire	void	0.60	1	1	Spit poll clou
Meadow Guardian	12	physical	lightning	ice	0.60	2	2	Mini boss vine + po barr

DESSERT DESERT — 5 mobs

Candy, sugar, sticky. Dessert mobs are sweet, sticky, and burn on contact. Element: **fire**. Mob HP = base HP × 1.3^ngp. Mob damage = listed dmg (flat; trash damage does not scale with NG). SPEED MAPPING: the Speed column is the SPEC abstract speed; the game maps it to px/frame: chaser 1.2 → 0.35, shooter 0.7 → 0.18, tank 0.45 → 0.08, spitter 0.6 → 0.14, elite 0.6 → 0.14 (before wave/NG scaling).

Name	HP	Element	Weak	Resist	Speed	Contact	Attack	Behavior	S
Gumdrop Runner	2	fire	ice	physical	1.20	1	0	Runs at the player, leaves a	S t p

								sticky trail.	
Candy Shooter	2	fire	void	fire	0.70	1	1	Fires candy shards from range.	C 1
Fudge Tank	8	fire	ice	physical	0.45	2	0	Slow, dense fudge armor.	A p c (u
Caramel Spitter	3	fire	ice	physical	0.60	1	1	Spits molten caramel.	C b ×
Dessert Guardian	12	fire	void	fire	0.60	2	2	Mini-boss: candy storm + syrup slam.	C (+ (s

FROST CAVES — 5 mobs

Ice, snow, wind. Frost mobs slow, freeze, and shatter. Element: **ice**. Mob HP = base HP × 1.3^ngp. Mob damage = listed dmg (flat; trash damage does not scale with NG). SPEED MAPPING: the Speed column is the SPEC abstract speed; the game maps it to px/frame: chaser 1.2 → 0.35, shooter 0.7 → 0.18, tank 0.45 → 0.08, spitter 0.6 → 0.14, elite 0.6 → 0.14 (before wave/NG scaling).

Name	HP	Element	Weak	Resist	Speed	Contact	Attack	Behavi
Snow Runner	2	ice	fire	lightning	1.20	1	0	Runs a the playe leave ice patch
Icicle Shooter	2	ice	physical	ice	0.70	1	1	Fires icicl bolts from range
	8	ice	fire	lightning	0.45	2	0	Slow, thick

Glacier Tank								glaci armor
Chill Spitter	3	ice	fire	lightning	0.60	1	1	Spits freez mist.
Frost Guardian	12	ice	physical	ice	0.60	2	2	Mini-boss: ice shard freez nova.

STORM PEAK — 5 mobs

Static storms crackle (3 storm cells shock you). Lightning mobs. Element: **lightning**. Mob HP = base HP × 1.3^ngp. Mob damage = listed dmg (flat; trash damage does not scale with NG). SPEED MAPPING: the Speed column is the SPEC abstract speed; the game maps it to px/frame: chaser 1.2 → 0.35, shooter 0.7 → 0.18, tank 0.45 → 0.08, spitter 0.6 → 0.14, elite 0.6 → 0.14 (before wave/NG scaling).

Name	HP	Element	Weak	Resist	Speed	Contact	Attack	Beha
Volt Runner	2	lightning	ice	lightning	1.20	1	0	Runs the play trai stat
Arc Shooter	2	lightning	physical	void	0.70	1	1	Fire bolt fron rang
Storm Tank	8	lightning	ice	lightning	0.45	2	0	Slow stor char armo
Static Spitter	3	lightning	ice	lightning	0.60	1	1	Spit stat char

Storm Guardian	12	lightning	physical	void	0.60	2	2	Mini bos ligh wav arc bar

HONEY MARSH — 5 mobs

Honey, mud, swamp gas. Marsh mobs stick, poison, and bog. Element: **void**. Mob HP = base HP × 1.3^ngp. Mob damage = listed dmg (flat; trash damage does not scale with NG). SPEED MAPPING: the Speed column is the SPEC abstract speed; the game maps it to px/frame: chaser 1.2 → 0.35, shooter 0.7 → 0.18, tank 0.45 → 0.08, spitter 0.6 → 0.14, elite 0.6 → 0.14 (before wave/NG scaling).

Name	HP	Element	Weak	Resist	Speed	Contact	Attack	Behavi
Bog Runner	2	void	lightning	physical	1.20	1	0	Runs a the playe throu mud.
Mud Shooter	2	void	fire	void	0.70	1	1	Fires mud balls from range
Mire Tank	8	void	lightning	physical	0.45	2	0	Slow, thick mire armor
Venom Spitter	3	void	lightning	physical	0.60	1	1	Spits venom
Marsh Guardian	12	void	fire	void	0.60	2	2	Mini-boss: bog wa + veno spray

THORNWOOD — 5 mobs

Brambles, vines, roots. Thorn mobs root, lash, and bleed. Element: **physical**. Mob HP = base HP × 1.3^ngp. Mob damage = listed dmg (flat; trash damage does not scale with NG). SPEED MAPPING: the Speed column is the SPEC abstract speed; the game maps it to px/frame: chaser 1.2 → 0.35, shooter 0.7 → 0.18, tank 0.45 → 0.08, spitter 0.6 → 0.14, elite 0.6 → 0.14 (before wave/NG scaling).

Name	HP	Element	Weak	Resist	Speed	Contact	Attack	Behavi
Vine Runner	2	physical	fire	ice	1.20	1	0	Runs at the player, vine-lashing
Thorn Shooter	2	physical	lightning	void	0.70	1	1	Fires thorn darts from range.
Bark Tank	8	physical	fire	ice	0.45	2	0	Slow, thick bark armor.
Nettle Spitter	3	physical	fire	ice	0.60	1	1	Spits nettle stings
Thorn Guardian	12	physical	lightning	void	0.60	2	2	Mini-boss: root spikes, thorn volley

VOID CASTLE — 5 mobs

Void, null, entropy. Void mobs phase, drain, and collapse. Element: **void**. Mob HP = base HP × 1.3^ngp. Mob damage = listed dmg (flat; trash damage does not scale with NG). SPEED MAPPING: the Speed column is the SPEC abstract speed; the game maps it to px/frame: chaser 1.2 → 0.35, shooter 0.7 → 0.18, tank 0.45 → 0.08, spitter 0.6 → 0.14, elite 0.6 → 0.14 (before wave/NG scaling).

Name	HP	Element	Weak	Resist	Speed	Contact	Attack	Behav

Shade Runner	2	void	lightning	physical	1.20	1	0	Runs a the playe phasi
Null Shooter	2	void	physical	void	0.70	1	1	Fires null bolts from range
Null Tank	8	void	lightning	physical	0.45	2	0	Slow, dense null armor
Entropy Spitter	3	void	lightning	physical	0.60	1	1	Spits entro
Void Guardian	12	void	physical	void	0.60	2	2	Mini-boss: null wave + entro barra

19. Future Candidates (SPEC — NOT IMPLEMENTED)

SPEC: These 35 archetypes (5 per terrain × 7 terrains: dasher, swarmer, lurker, bomber, support) are DESIGNED, not implemented. They are not in the game yet. Do not treat them as playable. No telegraphs defined (future work).

GREEN MEADOW — candidates

Role	Name	Behavior	Special	Sprite
SPEC dasher	**Breeze Dasher**	Dashes in 3-burst charges.	Dash burst: 3 quick charges, 60f rest.	Wind-swept grass tuft, speed lines.

SPEC swarmer	**Sproutling**	Spawns in groups of 3, weak.	Group spawn: 3 at once.	Tiny sprout, two leaves.
SPEC lurker	**Shade Lurker**	Hides in tall grass, ambushes when the player is near.	Ambush: first hit +1 dmg.	Shadow in grass, only eyes visible.
SPEC bomber	**Seed Bomber**	Explodes on death.	Death explosion: 1 dmg, radius 60px.	Round seed pod, fuse spark.
SPEC support	**Meadow Singer**	Sings: nearby mobs +20% speed.	Speed aura: radius 120px.	Flower-crowned figure, musical notes.

DESSERT DESERT — candidates

Role	Name	Behavior	Special	Sprite
SPEC dasher	**Licorice Dasher**	Dashes in 3-burst charges.	Dash burst: 3 quick charges, 60f rest.	Black licorice twist, red stripes.
SPEC swarmer	**Sprinkle Swarm**	Spawns in groups of 3, weak.	Group spawn: 3 at once.	Tiny rainbow sprinkle, one eye.
SPEC lurker	**Wrapper Lurker**	Hides under candy wrappers, ambushes.	Ambush: first hit +1 dmg.	Crinkled wrapper, peeking eyes.
SPEC bomber	**Jawbreaker Bomber**	Explodes on death.	Death explosion: 1 dmg, radius 60px.	Giant jawbreaker, cracked shell.
SPEC support	**Sugar Singer**	Sings: nearby mobs +20% damage.	Damage aura: radius 120px.	Lollipop-headed figure, sugar notes.

FROST CAVES — candidates

Role	Name	Behavior	Special	Sprite
SPEC dasher	**Frost Dasher**	Dashes in 3-burst charges.	Dash burst: 3 quick charges, 60f rest.	Pale frost wisp, ice trail.
SPEC swarmer	**Snowflake Swarm**	Spawns in groups of 3, weak.	Group spawn: 3 at once.	Tiny snowflake, six arms.
SPEC lurker	**Drift Lurker**	Hides in snowdrifts, ambushes.	Ambush: first hit +1 dmg.	Snow mound, only eyes visible.
SPEC bomber	**Frost Bomber**	Explodes on death.	Death explosion: 1 dmg + slow 1s, radius 60px.	Ice bomb, frozen fuse.
SPEC support	**Aurora Singer**	Sings: nearby mobs +20% speed.	Speed aura: radius 120px.	Aurora-veiled figure, shimmering notes.

STORM PEAK — candidates

Role	Name	Behavior	Special	Sprite
SPEC dasher	**Volt Dasher**	Dashes in bursts of lightning.	Dash burst: 3 charges, 60f rest.	Volt streak figure.
SPEC swarmer	**Spark Swarm**	Spawns in groups of sparks.	Group spawn: 3 at once.	Spark cluster.
SPEC lurker	**Storm Lurker**	Ambushes from the storm.	Ambush: first hit +1 dmg.	Storm shadow.
SPEC bomber	**Charge Bomber**	Explodes on death.	Death explosion: 2 dmg AoE.	Charged orb.
SPEC support	**Storm Caller**	Calls the storm for allies.	Aura: +20% speed (radius 120px).	Storm caller, lightning staff.

HONEY MARSH — candidates

Role	Name	Behavior	Special	Sprite
SPEC dasher	**Reed Dasher**	Dashes in 3-burst charges.	Dash burst: 3 quick charges, 60f rest.	Tall reed, bending stalk.
SPEC swarmer	**Mosquito Swarm**	Spawns in groups of 3, weak.	Group spawn: 3 at once.	Tiny mosquito, buzzing wings.
SPEC lurker	**Quagmire Lurker**	Hides in the bog, ambushes.	Ambush: first hit +1 dmg.	Bog mound, only eyes visible.
SPEC bomber	**Gas Bomber**	Explodes on death.	Death explosion: 1 dmg + poison 1 dmg/s x 3s, radius 60px.	Swamp gas balloon, green glow.
SPEC support	**Croak Singer**	Sings: nearby mobs +20% speed.	Speed aura: radius 120px.	Frog-headed figure, croaking notes.

THORNWOOD — candidates

Role	Name	Behavior	Special	Sprite
SPEC dasher	**Bramble Dasher**	Dashes in 3-burst charges.	Dash burst: 3 quick charges, 60f rest.	Tangled bramble, thorn spikes.
SPEC swarmer	**Seedling Swarm**	Spawns in groups of 3, weak.	Group spawn: 3 at once.	Tiny seedling, one leaf.
SPEC lurker	**Root Lurker**	Hides under roots, ambushes.	Ambush: first hit +1 dmg.	Root tangle, only eyes visible.
SPEC bomber	**Pod Bomber**	Explodes on death.	Death explosion: 1 dmg + bleed, radius 60px.	Thorn pod, bursting seams.
SPEC support	**Bloom Singer**	Sings: nearby mobs +20% damage.	Damage aura: radius 120px.	Bloom-crowned figure,

				petal notes.

VOID CASTLE — candidates

Role	Name	Behavior	Special	Sprite
SPEC dasher	**Phase Dasher**	Dashes in 3-burst charges.	Dash burst: phases through the player, 3 charges.	Phase wisp, shifting form.
SPEC swarmer	**Wisp Swarm**	Spawns in groups of 3, weak.	Group spawn: 3 at once.	Tiny wisp, purple glow.
SPEC lurker	**Rift Lurker**	Hides in rifts, ambushes.	Ambush: first hit +1 dmg + teleport behind player.	Rift tear, only eyes visible.
SPEC bomber	**Collapse Bomber**	Explodes on death.	Death explosion: 2 dmg + pull, radius 80px.	Collapsing void orb, imploding.
SPEC support	**Void Singer**	Sings: nearby mobs +20% damage.	Damage aura: radius 120px.	Void-veiled figure, null notes.

20. Status + Decisions (Game == Appendix)

GAME == APPENDIX (locked 2026-10-14). The appendix reflects the game AND the game reflects the appendix — they must match. If the appendix says X, the game must do X. If the game does Y, the appendix must say Y. Every item below carries a status so the reader knows what is live vs decided vs proposed. Status legend: **IMPLEMENTED** = in the game now. **QUEUED** = decided, not yet implemented. **DESIGNED** = spec / previews done, not yet in-game (preview-first: kyle hears it before it goes in). **PROPOSED** = not ordered — do NOT implement without kyle’s go-ahead.

20.1 Game == Appendix principle

IMPLEMENTED (as a rule). This appendix is generated from the canonical game file (banana-blaster.html) plus locked decisions recorded here. Where a decision is QUEUED, the appendix states the LOCKED rule and notes the game does not

yet have it. Where a design is DESIGNED, the appendix states the design and notes it is not yet in-game. NOTHING in this appendix is presented as implemented unless it is in the game.

20.2 Boss rotation bug + fix

IMPLEMENTED 2026-10-14 (verified via debug run, 0 frame errors). BUG (was in the game): BOSS_DEF has 22 bosses, BOSS_ORDER has 20, but the rotation used %15, so only the first 15 bosses ever spawned. **FIX (implemented):** %15 → %bossOrderRun.length (20); stale /15 cycle calculations → / bossOrderRun.length. **Pip & Pop stay OUT of the rotation** (kyle: they are the FINAL BOSS, not supposed to be in normal rotation — final boss only, wave 31, OHK fight). **Verified:** BOSS_ORDER = 20, no pip/pop, all 20 bosses spawn + attack + die, 0 frame errors. This supersedes the earlier "add Pip & Pop to BOSS_ORDER" plan (wrong — they are the final boss).

20.3 Boss reviews

COMPLETE. The 7 bosses were reviewed one by one (attacks, telegraphs, projectiles, loot, sprite, music, theme), fixed, and the 5 new bosses added to the rotation. **ORACLE = APPROVED (no changes).** Oracle (lightning, weak ice, resist void, tough 1.00, atkMul 1.15, pts 3000, sub "time is a gear"): 4 attacks (see §21) — REWIND (position rewind 2s), TIME STOP (freeze 1.5s + 5 gears), GEAR CRUSH (4 converging gears), CLOCK HANDS (2 rotating beams, 360° over 4s). Loot: count 3, high 1, lucky 0.10; signature Oracle Gear / Timepiece / Brass Key; sets: any (all bosses can drop any set — IMPLEMENTED). **Sugar Witch = APPROVED. Thorn Regent = APPROVED. Leviathan = APPROVED. Crystal Warden = APPROVED (maiden sprite). Pip & Pop = REVIEWED (final boss, not in rotation, attacks per spec, pick 2 of dropped items).**

20.4 All bosses loot = SAME (NG scales count)

IMPLEMENTED 2026-10-14. ALL bosses have the SAME loot counts at each NG level — no boss gets better loot than another. **NG+0: count 3, high 1, lucky 0.10. NG+1 / NG+2: count 4, high 1, lucky 0.10** (NG increases count to 4, high stays 1, lucky unchanged — flat, NOT +1 per NG level). **Pick: PICK 1 of the dropped items (unpicked are LOST) — reverses keep-all (kyle 2026-10-14, §20.25).** The 15 originals that had count 4, high 2, lucky 0.15 were REDUCED to the standard (3/1/0.10 at NG+0, 4/1/0.10 at NG+). Leviathan was already reduced to standard. **FINAL BOSS (Pip & Pop duo): player picks 2 out of the DROPPED items to keep (the rest disappear).** NG+0: duo drops 6 total (3 each) → pick 2 of 6. NG+1/NG+2: duo drops 8 total (4 each) → pick 2 of 8 (pick count stays 2, total scales with NG). The game now has uniform loot counts + NG scaling + final-boss

pick 2 (state 13). The boss cards below show the game's current loot (uniform).

20.5 Skybreaker — summon a PERIODIC STORM (locked)

IMPLEMENTED 2026-10-14 (verified via debug run, 0 frame errors). Skybreaker's lore says "Faster than the storm it summons" — its effect now matches the lore. **PERIODIC STORM (locked 2026-10-14, Option 1):** every **480 frames (8 seconds)** a storm strikes the arena. **Storm duration: 120 frames (2 seconds). Damage: 3 damage per 30-frame tick (4 ticks = 12 total per storm)** to enemies in the radius. **Radius: 120 px. Anchor:** random living non-boss enemy, else player position. **Element:** lightning; cause skystorm. **Visual:** translucent circle + rain streaks + flickering lightning bolt; shout "SKYBREAKER STORM!". kyle: "thats the best feels like a storm storms are uncontrollable." (Option 1 beats storm cloud (Option 2) and on-hit storm (Option 3) — storms are uncontrollable, so the storm is periodic/uncontrollable, not tied to the player's attacks.) The old +1 projectile effect is gone.

20.6 Voidtouched Dagger redesign — cooldown line (trap-line / tripwire)

IMPLEMENTED 2026-10-14. Replaces the old phased-ghost effect (weak: only 1 ghost mob type, phased 1/3 of the time — see §20.17). **NEW (locked):** cooldown line. **CD = 2s** (dagger = fast; mythic = powerful). Every 2s, the next shot leaves a line. **Line lasts 3s (180 frames).** Each monster crossing takes **4 dmg (once per line)**. **1 line at a time** (new line replaces the old). CD 2s + line 3s = line always available (re-place every 2s, feels fast). Lore updated to line-trap (IMPLEMENTED). ghostpierce has zero remaining occurrences in the game file.

20.7 Persistent stash cap 6 → 18

IMPLEMENTED 2026-10-14, Option 1. Persistent stash expands from 6 to **18 (6 × 3 — 3 rows/pages of 6)**. Holds ALL 18 set items (6 sets × 3pc) + 3 equipped = 21 total. **GOAL:** collect all 18 set items, try each set, finish NG+2 with each new set. Drives **REPLAYABILITY** (replay the save file until all set items are collected). Run stash stays cap 6 (unchanged — wiped at ADVANCE). The game now caps the persistent stash at 18 (MAX_PERSISTENT_STASH = 18, 6×3 layout). See also §13.

20.8 Music — per-boss + per-terrain + switching rule

IMPLEMENTED 2026-10-14. All 21 boss + 7 terrain v3 melodies/ harmonies wired into the procedural engine; switching connected (loop() calls Snd.music.auto every frame; state 13 → terrain; wave 31 overrides bossId to 'pip'); death/victory jingles wired (were never called). Each boss gets their OWN music (matching the boss's theme/element). Each terrain gets

its OWN battle loop (matching the terrain). This individualizes each map + boss. Total: 22 boss themes + 7 terrain battle loops + menu / victory / death. All JRPG-EDM, all original. **MUSIC SWITCHING (locked 2026-10-14):** during normal gameplay (exploring the map) → **terrain music** plays. Once the boss **spawns** → music switches to **that boss's theme**. When the boss **dies** → music switches back to the **terrain music**. (Switching connected 2026-10-14.) In-game music must switch correctly (terrain loop → boss theme when a boss spawns → victory / death) Live-browser audio checks still PENDING (not yet verified in a real browser)..

Music mapping — which music for which boss / terrain (locked 2026-10-14)

Boss / Terrain	Theme file	Version	Status
Vespa	boss-vespa-v2.wav	v2	APPROVED
Brinebeard	boss-brinebeard-v2.wav	v2	APPROVED
Moonhowl	boss-moonhowl-v2.wav	v2	APPROVED
Jester	boss-jester-v2.wav	v2	APPROVED
Honeybear	boss-honeybear-v3.wav	v3	APPROVED
Conductor	boss-conductor-v2.wav	v2	APPROVED
Frostbite	boss-frostbite-v2.wav	v2	APPROVED
Pyre Queen	boss-pyre-queen-v2.wav	v2	APPROVED
Ronin	boss-ronin-v3.wav	v3	APPROVED
Banana King	boss-banana-king-v2.wav	v2	APPROVED (Donkey Kong)
Dreadnought	boss-dreadnought-v2.wav	v2	APPROVED (Warhammer)

Gargantua	boss-gargantua-v2.wav	v2	APPROVED (metal)
Leviathan	boss-leviathan-v2.wav	v2	APPROVED (deep sea, lonely, scary)
Oracle	boss-oracle-v2.wav	v2	APPROVED (metronome)
Sugar Witch	boss-sugar-witch-v2.wav	v2	APPROVED (Sugar Plum Fairy)
Thorn Regent	boss-thorn-regent-v2.wav	v2	APPROVED (grandiose elf)
Pip & Pop	boss-pip-pop-v3.wav	v3	APPROVED (motif KEPT, twisted evil, A minor only, no mirror clash)
Librarian	boss-librarian-v3.wav	v3	APPROVED (Tibetan “mmmm”)
Crystal Warden	boss-crystal-warden-v3.wav	v3	APPROVED (softer, elegant, ice queen)
Starweaver	boss-starweaver-v3.wav	v3	APPROVED (more trancey)
Baron von Wobble	boss-wobble-v5.wav	v5	APPROVED (Kingdom Hearts: Chain of Memories)
Green Meadow (terrain)	terrain-green-meadow-v3.wav	v3	APPROVED (no motif)
Dessert Desert (terrain)	terrain-dessert-desert-v3.wav	v3	APPROVED (no motif)
Frost Caves (terrain)	terrain-frost-caves-v3.wav	v3	APPROVED (no motif)
Storm Peak (terrain)	terrain-storm-peak-v3.wav	v3	APPROVED (no motif)
Honey Marsh (terrain)	terrain-honey-marsh-v3.wav	v3	APPROVED (no motif)
Thornwood (terrain)		v3	APPROVED (no motif)

	terrain-thornwood-v3.wav		
Void Castle (terrain)	terrain-void-castle-v3.wav	v3	APPROVED (no motif)
Base pieces (battle / menu / victory / death)	(v1)	v1	APPROVED (keep the Banana Motif)

20.9 Music — style references + no shared motif in boss themes (except Pip & Pop) + v2/v3/v4/v5 approvals

DESIGN. Boss themes evoke specific style references (**ORIGINAL** — evoke the style, do NOT copy melodies; copyrighted pieces are references only, never copied). **Boss themes: NO shared motif** — each boss theme uses an ORIGINAL melody in its reference style (not a shared motif, not a variation of one) — **EXCEPT Pip & Pop** (the ONLY boss that keeps the Banana Motif, twisted evil — see below). References: **Vespa** = Flight of the Bumblebee (fast, frantic, chromatic runs, buzzing). **Brinebeard** = Nova Scotia sea shanty, pub by the sea (fiddle, accordion, stomping, call-and-response). **Moonhowl** = Beethoven (Moonlight Sonata vibes — piano triplets, strings, dramatic). **Jester** = clown circus song (calliope, waltz, off-kilter). **Honeybear** = Winnie the Pooh song (gentle, bouncy, warm, childlike). **Conductor** = Mozart / Bach IN OUR VERSION (classical orchestra, counterpoint, original melodies). **Frostbite** = coldness + ice (sparse, high, shimmering). **Pyre Queen** = Tchaikovsky flames (dramatic, fiery orchestral, 1812 vibes). **Ronin** = Battle on the Big Bridge energy (FF style reference — soulful + driving, evoke the energy, do NOT copy the melody). The remaining bosses keep v1. **Honeybear v3 APPROVED (2026-10-14):** softer, whimsical, soft + cuddly — no loud brass, soft flute + music box, heartbeat kick. (v2 rejected: too loud, not whimsical / soft / cuddly.) **Ronin v3 APPROVED (2026-10-14):** soulful + Battle on the Big Bridge energy — driving rock beat + soulful Hirajoshi melody, connected. (v2 rejected: disjointed.) Both are in the game (IMPLEMENTED 2026-10-14).

12 BOSS v2/v3/v4/v5 RE-DOS (2026-10-14, locked directions + approvals): Banana King v2 APPROVED (Donkey Kong inspired). **Dreadnought v2 APPROVED** (Warhammer feel). **Gargantua v2 APPROVED** (metal music). **Librarian v3 APPROVED** (Tibetan music, “mmmm” — throat singing, singing bowls, monastery bells, dungchen horns, no drums; v2 rejected: generic monks). **Starweaver v3 APPROVED** (more trancey — driving four-on-the-floor, layered pads with sidechain, denser arpeggios, supersaw lead, build + lift; v2 rejected: not trancey enough). **Baron**

von Wobble v5 APPROVED (Kingdom Hearts: Chain of Memories sounding — orchestral, piano-led, strings, adventurous, whimsical, Yoko Shimomura; keep the Wobble + variation; v2 rejected: didn't sound like a kingdom's band; v3 rejected: too boring, needs more variation; v4 rejected: make it more KH CoM sounding). **Crystal Warden v3 APPROVED** (sine melody + crystal bells kept, softer, more elegant, beautiful princess ice queen; v2 rejected: too harsh/loud). **Leviathan v2 APPROVED** (deep sea, lonely, scary). **Oracle v2 APPROVED** (more like a metronome). **Sugar Witch v2 APPROVED** (Sugar Plum Fairy — celesta, tinkling, sweet, ballet). **Thorn Regent v2 APPROVED** (grandiose elf music). **Pip & Pop v3 APPROVED** (Banana Motif KEPT — the ONLY boss that keeps it — twisted evil, A minor only, no mirror clash, no tritone shadows, evil but musical; v2 rejected: remove the mirror clash, keep the evil sound but make it sound good). All v2/v3/v4/v5: original melodies (except Pip & Pop's motif), NO Banana Motif (except Pip & Pop), keep key/mode/bpm/prog + distinct harmony. Implementation: IMPLEMENTED 2026-10-14 (all 12 re-dos wired into the procedural engine).

20.10 Music — terrain harmony + preview-first

DESIGN. The chords + notes must convey the theme (not just the instruments). Terrain v1 sounded too similar (same motif + same instruments + same scale). Terrain v2: distinct harmony per terrain — **Green Meadow** C major (Ionian) I–V–vi–IV; **Dessert Desert** A Dorian i–IV–i–VII; **Frost Caves** E Phrygian i–bII–i–VII; **Storm Peak** D Aeolian + diminished i–VII–VI–vi°; **Honey Marsh** G Lydian I–II–vi–IV; **Thornwood** B Harmonic minor i–iv–V–i; **Void Castle** F# Locrian i°–bII–i°–VII. **Terrain v3 APPROVED (2026-10-14): NO Banana Motif in terrain loops** (supersedes terrain v2 which kept the motif — v2 superseded, re-rendered as v3 with ORIGINAL melodies + distinct harmony). **Base pieces (battle / menu / victory / death) KEEP the Banana Motif** (unchanged — kyle only removed the motif from terrain + boss themes, except Pip & Pop). **PREVIEW-FIRST (locked):** render the music as audio files FIRST (before in-game implementation), post them so kyle can hear if each fits (the boss / terrain), THEN implement. Same flow as the 5 base pieces (render → listen → implement). **IMPLEMENTED 2026-10-14** (all approved themes wired into the procedural engine; see §20.8).

20.11 Sound — SFX IMPLEMENTED

IMPLEMENTED (verified 2026-10-14). SFX in-game (28 hooks, all verified, no gameplay numbers changed). Hit (throttled 1/80 ms), kill (elemental — see below), boss death, pickup (1/100 ms), player hit / death, set completion, NG prize, victory, vendor buy, equip, stash deposit / withdraw, ADVANCE, silenced, barrier break, poison tick, freeze, telegraph warning. **Elemental kill sounds** (by killing blow's damage type): physical = pop / thud, fire = crackle, ice = shatter,

lightning = zap, poison = sizzle, void = drain. **Bullet-hell rules:** no sound per bullet (player peas + enemy bullets are silent); one sound per volley; hit throttled to ≈1 per 80 ms; kill / pickup throttled to ≈1 per 100 ms; polyphony capped ≈12 (minor sounds skipped first, priority sounds always play); SFX short; important sounds briefly duck music (−3 dB). **Mute:** M key + visible HUD mute control; persists via localStorage key bb_sound_muted. Audio initializes only after user interaction. No errors when AudioContext is unavailable. Game file: 1,213,765 → 1,260,346 bytes (+46,581, sound only). Game file now 1,319,543 bytes after full implementation (2026-10-14).

20.12 Sound — rarity drop sounds IMPLEMENTED

IMPLEMENTED. When an item drops, play a sound by its RARITY tier, so the player HEARS how rare it is. Common = quiet tick. Uncommon = soft chime. Rare = brighter chime. Unique = 3-note chime. Mythic = 3-note + shimmer. Set = distinct fanfare (motif fragment). Signature = rarest sound (full motif + chime, loudest). If multiple items drop at once, play ONLY the highest-rarity sound. Throttle: one per 100 ms (60 ms debounce keeps the max tier). Sound only (no gameplay change).

20.13 ASCII mix

IMPLEMENTED (locked). Keep the mix — ASCII sprites (fillText, Courier New) + vector FX / UI (telegraphs, projectiles, particles, HUD). “It’s a good mix.” Do NOT make FX / HUD ASCII (pure ASCII rejected). Telegraphs stay vector (hitbox rule: telegraph draw must match the hitbox exactly).

20.14 Post-cohort fixes (IMPLEMENTED 2026-10-14)

IMPLEMENTED 2026-10-14 (verified via debug run: 0 frame errors, 0 warnings, syntax OK, save-load PASS). (1) Elite spawn gate: **Wave 3** — already satisfied (variantChance already gated elite at Wave 3; no change needed). Final gates: chaser / shooter / spitter Wave 1, tank Wave 2, elite Wave 3. (2) DrawEquip Option C: redraw ALL 6 equipment buttons (SELL / DROP / SWAP / DELETE / DEPOSIT / WITHDRAW) after the screen clear, uniformly. **NOTE: visual + functional check in a real browser still PENDING.** (3) Signature items ONLY from the dedicated 5% roll; the normal unique/set rarity path EXCLUDES boss signature items; total signature-item chance ≈5%. (4) Persistent stash 6 → 18 (see §20.7). (5) Skybreaker storm (see §20.5). (6) Dagger redesign (see §20.6). (7) Boss rotation fix (see §20.2). (8) Boss attacks theme-specific (see §20.19). (9) All bosses drop any set (see §20.21).

20.15 Endless mode (PROPOSED — NOT ordered, DEFERRED)

PROPOSED. IF endless mode is added (after the game works), difficulty scales ONLY by monsters increasing HP + damage (that's it). Item stars do NOT increase (stars cap at 2 — the NG+2 max, no endless gear power creep). The player must rely on skill / build. Priority: fun first, extra polish later (endless mode is pointless if the base game isn't fun). Do NOT implement without kyle's go-ahead.

20.16 Mythic items (9, verified)

IMPLEMENTED (all 9, incl. the 2 redesigns, 2026-10-14). One per weapon type (8 hands + neck). **Worldsplitter** (sword, shockwave: on-kill → small shockwave). **Aegis of Dawn** (shield, dawnshield: once per wave blocks a lethal hit). **Voidtouched Dagger** (dagger, LINE TRAP — IMPLEMENTED, see §20.6). **Stormcaller Wand** (wand, stormcall: every 10s lightning strikes a random enemy for 1 dmg). **Heart of the Mountain** (neck, thorns: when hit, nearest enemy takes 1 dmg). **World Staff** (staff, knockback). **Worldpiercer** (spear, worldpierce: infinite pierce). **Skybreaker** (javelin, SUMMON A PERIODIC STORM — IMPLEMENTED, see §20.5). **Doombringer** (crossbow, doom: on-kill, nearby enemies take 1 dmg). **NO mythic item summons a minion** (summoning is enemy-side only: bosses MINION CALL, elites Summoner / Horde).

20.17 Ghost

IMPLEMENTED (mechanic). 1 ghost mob type (the 'ghost', Wave 4+, spawn-pool weight 10, count 3, phased 1/3 of the time). Plus void enemies with skillPhase (also phase 1/3 of the time). When phased: semi-transparent, shots pass through (without the Dagger redesign), set skills skip, damage halved. The old phased-ghost effect was weak (only matters vs phased enemies, a minority) — hence the Dagger redesign (§20.6, IMPLEMENTED).

20.18 Polish features

IMPLEMENTED 2026-10-14 (verified via debug run, 0 frame errors). Ordered 2026-10-14, required before hosting: (1) Improved onboarding (drop-skill farming, vendor, sets / forms, run stash, persistent stash, ADVANCE vs REPLAY, keep-three + star-one prize) + prompted-moves primer (4-step sequential gating, dummy HP 1→999). (2) In-game tooltips with formulas (real values, not vague descriptions). (3) Pause / settings menu (P, RESUME/MUTE/RESTART/QUIT). (4) Death + victory end screens (stats + replay). (5) Boss intros (name banner + health bar) + boss health bars in HUD. (6) One-click metrics export (deaths, run time, damage, build, wave reached, first vendor purchase wave; exportMetrics() JSON download). (7) **JUICE PACK (IMPLEMENTED 2026-10-14):** game-feel package — hit

flash (enemy flashes on hit), damage numbers (pooled, 40; pop on hit), small shake (kill-only, 6/3 — per-hit shake at pea fire rates would be constant vibration), pickup sparkle (pooled, 120), pop (visual ring on pickup/kill). Makes the game feel tight and precise (it should only feel bad when an enemy casts something or a terrain debuff hits). Juice pack does NOT change gameplay numbers (damage, HP, cooldowns, costs) — it ADDS feel, it does not replace anything. **Bug found + fixed during implementation:** the pea-hit damage-number call referenced block-scoped dmgType (ReferenceError on every shield-down pea hit, 38 frame errors, blocked wave completion) — fixed to spawnDmgNum(..., z.lastDmgElem || 'physical'). **NOTE: live-browser checks still PENDING** (pause menu, health bar, metrics download, death/victory screens, equipment buttons, audio) — not yet verified in a real browser.

20.19 Boss attacks — ALL 22 theme-specific

IMPLEMENTED 2026-10-14 (verified via debug run: every boss's attack cycle forced atkT = 1 → windup → resolve, 0 frame errors). LOCKED (kyle, Option 2, 2026-10-14): ALL 22 bosses get theme-specific attack MECHANICS (not just renamed generic attacks). OLD attacks were generic (fan / ring / slow / beam / freeze / meteors) with theme paint (name + color + shape). NEW (in-game now): each boss's attacks DO something that only that boss does — the mechanic IS the theme. Full spec in §21: 84 attacks (21 groups × 4; Pip & Pop share 4 duo attacks = 22 bosses), each with mechanic + telegraph + projectile + damage + cooldown / duration + theme link, plus the 5 attacks that could not be pure-damage theme-specific + implementation notes. **Leviathan per kyle's picks (overrides the spec's earlier Leviathan section):** Tidal Surge (KEEP), Whirlpool (KEEP), Ink Cloud (NEW, replaces Abyssal Spit — "ink cloud is cool"), Crushing Depths (KEEP, signature). Abyssal Spit = OUT ("stupid"). The generic attacks are gone (all 84 theme-specific attacks in execBossAtk).

20.20 Appendix status

This update adds everything above (all 2026-10-14 decisions). The generator (generate_appendix.py + spec_new_content.py) was updated, not just the HTML — re-running the generator reproduces this appendix. **PDF regenerated 2026-10-14 from the current HTML** (Gap 10 fix): the old PDF (47 pages, predating §20/§21) was backed up to banana-blaster-visual-appendix-stale-20261014.pdf and replaced with a fresh render of the current HTML (136 pages). **Re-rendered again 2026-10-14 post-implementation** (after the appendix was updated to IMPLEMENTED for all queued items; the pre-implementation PDF backed up to banana-blaster-visual-appendix-20261014-preimpl.pdf). **Render path (repaired 2026-10-14):** headless Chromium --print-to-pdf is BROKEN in this environment (the meta-chromium build hangs —

even on a trivial page — and no chromium binary is on PATH; the earlier >220s hang was Chromium itself, not the HTML). Working path: **WeasyPrint 70** (venv /tmp/pdf-venv) + print-override stylesheet print-override.css (saved in the banana-blaster dir). **Why the override is needed:** the HTML's @media print rule flips text to black but leaves the dark .card / pre / canvas backgrounds dark — without the override, card text and sprite pre blocks render black-on-black (invisible) in ANY print renderer (WeasyPrint or Chromium). The override completes the print intent (white backgrounds, black text). **Known WeasyPrint limitation:** the 134 JS-drawn <canvas> telegraph-shape mini-diagrams render blank (WeasyPrint does not execute JS) — each shape is also stated in text ("Telegraph shape: X"), so no information is lost. Re-render command: weasyprint HTML + print-override.css → PDF (see the override file's header comment). Verification: no stale Wave-4 elite-gate language, no signature double-count language, no hardest-boss loot-exception language, no old Dagger effect, no legacy-hp-as-HP presentation, no shared-motif-in-boss-themes language, no old generic boss attack descriptions (§21 has the full theme-specific spec, incl. kyle's Leviathan picks), no per-boss "associated set" (all bosses drop any set, §20.21). Game == appendix: the appendix accurately describes the game + clearly marks QUEUED / DESIGNED / PROPOSED items as not-yet-implemented. **Updated again 2026-10-14 for the 6-fix playtest update (§20.25); updated again 2026-10-14 for the stash/ locked/perma-stash fixes (§20.26); PDF re-rendered from the current HTML.**

20.21 All bosses drop any set

IMPLEMENTED 2026-10-14. ALL BOSSES DROP ANY SET. The "associated set" field is **REMOVED** — no boss is restricted to one set. ANY boss can drop items from ANY of the 6 sets (Moonfang, Frostbound, Tempest, Sunfire, Voidtouched, Swarm). **Farming:** no targeted farming by element — any boss can drop any set piece (set collecting is random, not element-gated). **Clarification:** the signature roll is NOT affected — the 5% roll still drops THAT boss's own signature items (e.g., Oracle Gear from Oracle). Signature items themselves belong to sets (Oracle Gear is a Tempest piece) — that is the ITEM's set, not a boss restriction. The "any set" rule governs the normal set-rarity drops.

20.22 Audit fixes (2026-10-14) — Crystal Warden maiden sprite, sprites match intro, tutorial, terrain debuff, WASD, Wave 0

IMPLEMENTED 2026-10-14 (verified via debug run, 0 frame errors). Found by the 2026-10-14 audit (items said/promised but missing from the appendix/implementation): (1) **Crystal Warden maiden sprite:** the new sprite is a beautiful

crystallized maiden (tiara, soft face, bodice gem, flaring crystal gown). IMPLEMENTED: maiden art in BOSS_DEF + regenerated DENSE_CRYSTALWARDEN (44×32, 1408 cells). (2) **Sprites match intro screen:** the sprites (BOSS_DEF art) must match the intro/title screen sprites. IMPLEMENTED: 22 BOSS_DEF art arrays + 22 DENSE_* entries present; Level Brief uses drawDense. (3) **Prompted-moves tutorial:** the tutorial teaches through one prompted action at a time (Rubics Solver pattern); each prompt advances only after the player performs it. Sequence: movement, shooting, pickup/drop interaction, weakness use. IMPLEMENTED: prompted-moves primer (4-step sequential gating, dummy HP 1→999). (4) **Terrain debuff visual feedback:** replace red-text-only debuff presentation with: ground glow + particles, player tint, persistent debuff icon while active. (The bug: red text says the debuff is happening but there is no animation.) IMPLEMENTED: ground glow + particles + player tint + persistent HUD icon. (5) **WASD + mouse precision principle:** controls must feel precise/ responsive; should only feel bad when an enemy casts something or a terrain debuff hits. Verified BY CODE (direct input, no smoothing). Real-mouse live check still PENDING. (6) **Wave 0 grants the drop skill (instant fun):** Wave 0 grants the drop skill; skipping intro + Wave 0 means missing the grant. IMPLEMENTED: Wave 0 grants the drop skill. (7) **ATK_TELL §21 shouts (87 keys):** IMPLEMENTED. (8) **Dagger lore:** updated to line-trap (IMPLEMENTED). (9) **Cartridge rules applied to juice pack:** pooled dmgNums/sparkles, no hot-loop allocations (IMPLEMENTED).

20.23 Deferred gaps (audit 2026-10-14, kyle’s calls)

DEFERRED (not in current implementation). Audit gaps kyle deferred 2026-10-14 ~20:18 EDT: **Gap 3 (player sprites redesign) = WAIT.** Detailed, beautiful screens + redesigned player sprites; kyle chose both male and female character options; Ness-like (cap/striped-shirt/backpack/shorts/sneakers energy) without copying. **Gap 4 (intro/lore playable setup) = DO LATER.** Short playable setup: character outside at night looking for “them,” trips and falls into the hole, gradually discovers sister is there, must rescue her and get back. Keep intro short. (“Them” = sister alone vs sister + someone else — unanswered.) **Gap 5 (touch controls, mobile) = DO LATER.** After desktop/hosting. **Gap 6 (boss hitboxes → shape-based inShape) = DO LATER.** Verify boss hitboxes use inShape; if not, implement. **Gap 7 (Ned sprite redesign) = DO LATER.** kyle decides the new sprite; current is ugly. (Gap 3 vs Gap 7: may be one item — kyle to confirm whether the player-sprites redesign and the Ned sprite are the same.) **Gap 10 (PDF stale + PDF render path) = DONE.** Repaired + regenerated 2026-10-14 (WeasyPrint + print-override.css); re-rendered again post-implementation. **Gaps 1+2 (prompted-moves tutorial, terrain debuff visual feedback) = DONE (IMPLEMENTED 2026-10-14, see §20.22).**

20.24 Debug-run fixes (2026-10-14) — .x / .phase / .phased errors + optimization

IMPLEMENTED 2026-10-14 (verified via debug run, 0 frame errors). .x / .phase / .phased errors — FIXED. Root cause: zombie-loop index desync from nested multi-splices (killZombie chains: Doombringer gear, Volatile → explode(), shockwave gear → explode(); one iteration can splice 2+ zombies, so a backward-loop index points past the shrunken array, zombies[_zi] is undefined, and the next property read throws). Fix: re-sync clamp BEFORE the binding in all 8 killing loops (if (_zi >= zombies.length) _zi = zombies.length - 1; if (_zi < 0) break; const z = zombies[_zi];). Verified: desync stress (doom + shockwave gear, 12 + 8 volatile 1-HP poisoned zombies, direct explode() into a cluster, 420 frames) = 0 frame errors; full debug run = 0 frame errors. **Known caveat (pre-existing, non-crashing):** a below-index splice can still cause one already-processed zombie to be visited twice in a frame (double turn) — left alone to avoid a riskier refactor. **Optimization:** removed provably dead code — callBoss() function + the callboss dispatch branch (no boss seq contains 'callboss'), boss-branch hiveT check (bosses never carry hiveT). Cartridge rules followed (pooled objects, in-place loops, no hot-loop allocations, NOT minified, comments NOT stripped). Full debug run re-run after optimization: ALL PASS, 0 errors.

20.25 Playtest fixes (2026-10-14) — tutorial timer, Wave 1 skill, boss HP x3, pick 1 of 3, readability, 1280x960

IMPLEMENTED 2026-10-14 (kyle playtest, 6 items; verified: syntax OK + debug run ALL PASS, 0 frame errors). (1) **Tutorial:** primer NO TIMER — ends only when all 4 steps (move, shoot, pick skill, hit) are done. ESC / SKIP still exit. (2) **Wave 1 (tutorial wave) skill:** exactly ONE add drops a guaranteed random drop skill (first non-boss kill); after that, Wave 1 adds never drop skills. Hearts/currency unchanged. (3) **Boss HP x3:** bossHPForWave x3 — Wave 1 = 60, Wave 12 = 225, Wave 15 = 315, Wave 30 = 765 (before 0.9375 x tough). Spawned HP = bossHPForWave x 0.9375 x tough. (4) **Boss loot PICK 1 of 3:** player picks 1 of the 3 dropped items (NG+0) / 1 of 4 (NG+1/2); unpicked are LOST. Reverses keep-all. Final boss (Pip & Pop) stays pick 2. (5) **Readability:** vendor labels 9→12px, vendor sublabels 8→10px, tooltips 11→13px, levelComplete card names 11→14px, HUD bumps; one message layer (floats + banner hidden while the pick screen is open). (6) **Display standard 1280x960:** wrapper max-width 680→1320px, game displays ~1296x972; logic stays 640x480, CSS scales 2x, mouse math already scales via getBoundingClientRect.

20.26 Stash warning, locked pickup, perma-stash screen (2026-10-14)

IMPLEMENTED 2026-10-14 (kyle playtest, 3 items; verified: syntax OK + debug run ALL PASS + targeted sim ALL PASS). (1) **Stash warning flashing forever — FIXED.** Root cause: loot magnet dragged full-stash loot onto the player, and collectDrop shouted INVENTORY FULL (6/6)! EVERY frame while the blocked pickup sat on the player (fresh 90-frame shout per frame = never disappeared). Fix: blocked-collect shouts throttled to 1 per 120 frames (same throttle covers SET SKILLS ACTIVE), AND loot stops magnetizing onto the player while the stash is full (stays where it dropped until the player makes room). (2) **Locked drop skills never disappearing — FIXED.** Root cause: the old 40px hover ring pushed locked skills back out, but PICKUP_RADIUS is 26, so locked skills could NEVER touch the player and hovered forever. Fix: hover ring removed; magnet pulls locked skills to contact and the caught handler deletes them (first-time SKILLS LOCKED! hint kept). (3) **Perma-stash screen — NEW (state 14).** After finishing the game (victory), BEFORE starting NG / NG+: ADVANCE opens the persistent-stash screen (P3 REPLAY NG++ too). Run stash ↔ persistent stash (click to deposit / withdraw, caps 6 / 18); DONE goes to the prize (ADVANCE) or replays NG++ (P3); BACK / ESC returns to victory. Equipped items stay in the prize pool (deposit/withdraw is run-stash only).

21. Theme-Specific Boss Attacks (IMPLEMENTED)

Status: IMPLEMENTED 2026-10-14 (verified via debug run: every boss's attack cycle forced atkT = 1 → windup → resolve, 0 frame errors). This section REPLACES the old generic boss attack descriptions (fan/ring/slow/beam/freeze/meteors) — the appendix describes the theme-specific attacks now in the game. The generic attacks are gone.

Source: Current regulars are generic (bFan/bRing/bMeteor/bSlow/bCracks/bBurn/spin, recolored per boss). Current signatures are a mix — several are already theme-specific (Brinebeard Tidal Wave, Moonhowl Blood Moon, Vespa Hive, Dreadnought Siege, Pip Betrayal, Librarian Silence) and are kept/upgraded; the rest are rebuilt.

Rules kept: 4 attacks per boss (3 regular + 1 signature); telegraph shape = BOSS_SHAPE[bossId] (hitbox matches draw); damage = atkMul × base boss projectile damage (DoT ticks = 0.5× base × atkMul per tick); windup = 180f (3s, NG −60f in P2/P3 via ngTeleT); rotation cadence = boss attack timer. **Stats unchanged** (tough/atkMul/element/weak/resist untouched —

mechanics only). Total: **84 attacks** (21 groups × 4; Pip & Pop share 4 duo attacks = 22 bosses).

RONIN (physical, iso, atkMul 1.1)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme li
A1 IAIDO CUT	Single instant slash line — no projectiles; damage if player is on the line at resolve	Iso line, len 320, w 40, t 180f	None (melee line)	atkMul × base (1 hit)	Rotation cadence	Iaido = perfect cut, no projecti
A2 SWORD SAINT	Marks player (circle), then dashes to that spot and slashes (line)	Iso circle r 60 at player, t 150f → iso line len 200	None (dash + slash)	atkMul × base	Rotation cadence	Precise tracking strike
A3 HONOR DUEL	Iso arena (r 150) around player for 4s; Ronin dashes in and slashes twice; leaving early ends it	Iso circle r 150, t 180f, dur 240f	None (2 slashes)	atkMul × base per slash (2 hits)	Rotation cadence; arena 240f	The duel bounded
SIG SEPPUKU SLASH	Dashes THROUGH player (past), leaving a delayed iso line that detonates after 1s for a second hit	Iso dash line + delayed iso line, t 180f windup + 60f delay	None (dash + delayed line)	atkMul × base × 2 hits	Rotation cadence	Desperat final sl — last honor

PYRE QUEEN (fire, heart, atkMul 1.2)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Them link

A1 THRONE OF FLAMES	Heart ring of fire closes in (r 220 → r 60 over 180f), then burns	Heart ring, r 220→60, t 180f	None (zone)	atkMul × base if caught at close	Rotation cadence	The thro clos in
A2 BURNING CROWN	6 fire orbs orbit Queen (r 90, 240f), then fly outward at player (homing)	Heart telegraph r 90 around boss, t 180f (orbs visible in orbit)	6 homing fire orbs, heart, spd 0.25	atkMul × base per orb	Rotation cadence; orbit 240f	The crow orbi then stri
A3 INFERNO WAVE	Wall of fire (heart band, w 80, full arena height) sweeps horizontally from Queen's side	Heart band, t 180f, sweep 1.1px/f	None (sweeping wall)	atkMul × base if touched	Rotation cadence	A wa of f
SIG ROYAL IMMOLATION	Sets whole arena floor on fire (fire patches, 4s) + heart ring (12 shots)	Full-arena fire telegraph (heart outline), t 180f	12 heart shots, ring	atkMul × base per patch tick (0.5×/ tick) + per shot	Rotation cadence; patches 240f	The aren burn the Quee ulti

FROSTBITE (ice, equilateral, atkMul 0.9)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration
A1 GLACIER CRUSH	3 ice blocks (equilateral) fall at player's position, staggered 30f	3 equilateral triangles r 50, t 180f	Falling blocks (equilateral)	atkMul × base per block	Rotation cadence
A2 PERMAFROST	Ground freezes in a growing equilateral	Equilateral triangle, r 60→160, t	None (zone)	atkMul × base on crack	Rotation cadence; grow 240f

	triangle (r 60 → r 160 over 240f) — slows inside; cracks for damage on completion	180f, dur 240f			
A3 AVALANCHE	Wave of ice (equilateral band, w 100) sweeps arena from top, pushing player down + damage	Equilateral band, t 180f, sweep 1.0px/f	None (sweeping wave)	atkMul × base	Rotation cadence
SIG GLACIER	3 big slow zones (equilateral, r 120, 4s) + a crack (equilateral, r 100) at center — keep current bGlacier, reskinned	3 equilateral slow zones + crack, t 180f	None (zones + crack)	atkMul × base (crack) + slow	Rotation cadence; zones 240f

CONDUCTOR (lightning, hexagon, atkMul 1.3)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration
A1 BATON STRIKE	On the beat (1s intervals), lightning bolt strikes player's position — 3 strikes, one per beat	Hexagon r 50, t 60f (short, rhythmic) × 3	Lightning bolts (hexagon)	atkMul × base per bolt	Rotation cadence; 3 beats
A2 ORCHESTRA HIT	All "instruments" fire at once — hexagon ring (12) + fan (5) simultaneously	Hexagon ring telegraph r 150, t 180f	12 ring + 5 fan, hexagon	atkMul × base per shot	Rotation cadence
A3 CRESCENDO	Attacks accelerate —	Hexagon pulse	None (buff)	None (accelerates	

	next 3 attacks fire at 2× cadence for 10s (buff, no direct damage)	around boss (visual metronome speeding up), t 180f		other attacks)	Rotation cadence; buff 600f
SIG FINALE	Massive coordinated barrage — 3 hexagon rings (12 each) + 2 fans (5 each) in sequence over 3s, on the beat	Hexagon rings, t 180f	36 ring + 10 fan, hexagon	atkMul × base per shot	Rotation cadence; 3s sequence

HONEYBEAR (physical, rect, atkMul 0.9)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme link
A1 HONEY POOL	Rect pool of honey (r 100) at player — sticky (0.5× speed), lasts 4s, then bursts (damage if still inside)	Rect pool r 100, t 180f, dur 240f	None (pool)	atkMul × base on burst	Rotation cadence; pool 240f	Honey traps you
A2 HONEY DRIP	Honey drips fall from above (5 drips, staggered) — each sticks to player (slow stack, 1s each) + damage on landing	5 rect r 30, t 180f	Drips (rect)	atkMul × base per drip + slow stack	Rotation cadence	Honey rains down
	Honeybear dashes to	Rect r 80, t	None (grab)	atkMul × base		

A3 BEAR HUG	player and grabs (rect r 80) — if player still inside at resolve, held (1s, can't move) + damage; moving breaks it early	180f, hold 60f		on grab	Rotation cadence; hold 60f	The bear hug
SIG SNACK TIME	Honeybear eats a snack (heals 10% HP) then attacks — honey ring (10) + honey pool at player	Rect ring r 150 + rect pool r 100, t 180f	10 rect shots	atkMul × base per shot + burst; heal 10% HP	Rotation cadence	Snack time — heals, then attacks

JESTER (void, right, atkMul 1.0)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme link
A1 JOKE BOMB	Gift box at player's position — looks like a pickup, then explodes (r 80)	Right triangle r 50, t 180f, then explosion r 80	None (gift → explosion)	atkMul × base	Rotation cadence	The gift is a bom
A2 FUNHOUSE MIRROR	2 mirrors (right-triangle walls) appear — Jester's next fan reflects off them (shots bounce,	2 right-triangle mirrors, t 180f, dur 240f	Fan (4) + reflected fan (4)	atkMul × base per shot	Rotation cadence; mirrors 240f	Mirrors reflect the joke

	doubling the fan)					
A3 PIE IN THE FACE	Pie (fast) flies at player — on hit: blinded (screen flash) + slowed 2s + damage	Right-triangle telegraph at player, t 150f	1 pie (right), spd 0.35	atkMul × base + blind/slow 120f	Rotation cadence	Pie in the face
SIG PUNCHLINE	Fake-out — telegraphs a big attack (r 150) that does NOTHING, then instantly attacks from behind player (fan 4 + ring 8)	Fake right triangle r 150 (no damage) + real fan/ring	4 fan + 8 ring, right	atkMul × base per shot	Rotation cadence	The punchlin — fake-out is the setu

MOONHOWL (physical, crescent, atkMul 1.2)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	The lin
A1 MOONLIGHT HOWL	Howls — buffs self (speed +20% for 5s) + summons 1 wolf (sprinter)	Crescent r 150 around boss, t 180f	None (buff + summon)	None (buff)	Rotation cadence; buff 300f	The cal pac emp
A2 LUNAR POUNCE	Leaps at player — lands with a crescent shockwave (r 100)	Crescent r 70 at player, t 150f, then dash 70f	None (leap + shockwave)	atkMul × base on landing	Rotation cadence	The pou
A3 PACK HUNT	3 wolves (sprinters, 1.2× speed) spawn and	Crescent telegraphs at spawn	None (summons)	Wolf contact = atkMul × base	Rotation cadence	The hun

	chase player	points, t 180f				
SIG FULL MOON	Transforms (bigger, faster — speed +40% for 10s, red tint) + crescent ring (16 shots)	Crescent ring r 150, t 180f	16 crescent shots	atkMul × base per shot; transform buff 600f	Rotation cadence; transform 600f	The moo the unl (ke Blo Moo cor tin rin wol

BRINEBEARD (ice, triangle, atkMul 1.1)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Them link
A1 TIDAL WAVE	Water band (triangle, w 100) sweeps horizontally from boss's side (smaller than signature)	Triangle band, t 180f, sweep 1.1px/f	None (sweeping band)	atkMul × base if touched	Rotation cadence	The tide
A2 ANCHOR DROP	Anchor (triangle, r 60) falls from above at player's position, then crashes (damage + shockwave ring r 80)	Triangle r 60, t 180f	Anchor (triangle)	atkMul × base (anchor) + atkMul × base (shockwave)	Rotation cadence	The anch drop
A3 FROZEN TIDE	Ground freezes in waves — 3 triangle slow zones (r 90) ripple outward from boss (staggered 30f), each 3s	3 triangle slow zones, t 180f	None (zones)	Slow + atkMul × base if still inside at expiry	Rotation cadence; zones 180f each	The froz tide

SIG SEA SHANTY	Sings — sound wave (expanding triangle ring, r 60 → r 250 over 120f) damages + knocks back, then a big tidal wave (triangle band, w 150) sweeps	Triangle ring + band, t 180f	Sound wave + tidal wave	atkMul × base per hit (wave + shanty)	Rotation cadence; wave 120f	The shan — th sea sing then stri

VESPA (lightning, octagon, atkMul 1.3)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme link
A1 SWARM	3 bees (homing, octagon, spd 0.3) chase player for 5s or until killed	Octagon telegraphs at spawn, t 180f	3 homing bees	atkMul × base per bee sting	Rotation cadence; bees 300f	The swarm chases
A2 STING	Fast stab — Vespa dashes at player (line, t 120f — fast) and stings (damage + poison tick 1 dmg/s for 3s)	Octagon line, len 300, w 40, t 120f	None (dash + sting)	atkMul × base + poison 1/s × 3s	Rotation cadence	The sting
A3 HIVE	Spawns a hive (octagon, r 50, 10s) that spawns 1 bee every 2s — hive	Octagon hive r 50, t 180f	Bees (homing)	atkMul × base per bee	Rotation cadence; hive 600f	The hive spawns bees — kill it

	can be destroyed (3 hits)					
SIG ROYAL STING	Big sting that tracks — charges, then a large homing stinger (octagon, r 40, spd 0.4) tracks player for 3s	Octagon r 100, t 180f	1 big homing stinger	atkMul × base (single big hit)	Rotation cadence; track 180f	The royal sting — the queen's strike (replaces HIVE signature; Hive → A3)

WOBBLE (void, trapezium, atkMul 1.0)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Th
A1 WOBBLE	Wobble wobbles — next fan curves (shots wobble sinusoidally, amplitude 30px)	Trapezium fan telegraph, t 180f	5 curving shots	atkMul × base per shot	Rotation cadence	Th wo
A2 TIPSY CHARGE	Charges at player but stumbles (path zigzags) — on hit: damage + knockback; on miss: Wobble falls (stunned 1s, vulnerable)	Trapezium zigzag path, t 180f	None (charge)	atkMul × base on hit; miss = Wobble stunned 60f	Rotation cadence	Ti — if pu it
A3 OFF-BALANCE	Falls over — then gets up with a shockwave (trapezium ring, r 120 → r 200, 1s)	Trapezium r 120, t 180f	None (shockwave)	atkMul × base on shockwave	Rotation cadence; shockwave 60f	Fa sh ge

SIG NOBLE STUMBLE	Random-direction burst — stumbles and fires a burst in a RANDOM direction, then a second burst in the opposite direction	2 trapezium bursts (random + opposite), t 180f	2 bursts (6 shots each, trapezium)	atkMul × base per shot	Rotation cadence	Th st un

LIBRARIAN (ice, parallelogram, atkMul 1.0)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldow Duratic
A1 SHUSH	Silence zone (parallelogram, r 90) at player — if inside at resolve, set skills + drop-skill effects locked 3s	Parallelogram r 90, t 180f, lock 180f	None (lock)	None (lock)	Rotatic cadence lock 18
A2 BOOK STORM	Books fly at player — fan of 6 books (parallelogram, spread 2.4, spd 0.25)	Parallelogram fan telegraph, t 180f	6 books	atkMul × base per book	Rotatic cadence
A3 DUSTY TOME	Large book (parallelogram, r 60) opens and fires a beam (len 400, r 30)	Parallelogram beam, len 400, r 30, t 150f	Beam (parallelogram)	atkMul × base	Rotatic cadence
SIG OVERDUE	Debt — player takes damage over time (0.5×/tick, 1 tick/s, 5s) UNLESS they move (moving pauses the debt; standing still accrues it)	Parallelogram debt mark on player, t 180f	None (DoT)	0.5× base per tick × atkMul if still; paused while moving	Rotatic cadence debt 30

GARGANTUA (physical, square, atkMul 1.2)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	The
A1 MOUNTAIN STOMP	Stomps — shakes screen + damages in radius (r 150)	Square r 150, t 180f	None (stomp)	atkMul × base in radius	Rotation cadence	The mou sto
A2 AVALANCHE	Rocks fall — 5 rocks (square, r 40) fall at player's position, staggered 30f	5 square r 40, t 180f	Falling rocks (square)	atkMul × base per rock	Rotation cadence	The ava
A3 BOULDER THROW	Throws a huge rock (square, r 50, spd 0.3) at player — rock bounces once	Square rock path, t 180f	1 boulder (square, bounces once)	atkMul × base per hit (2 possible hits)	Rotation cadence	The thr
SIG UNSTOPPABLE	Immune to knockback + walks through player (contact damage) for 5s + square ring (8 shots)	Square ring r 150, t 180f	8 square shots	atkMul × base per shot + contact	Rotation cadence; unstoppable 300f	Uns — t mou wal thr (re BOU ROL Bou Thr

DREADNOUGHT (fire, arrow, atkMul 1.3)

Attack	Mechanic	Telegraph	Projectile	Damage	Cool Durat
A1 MARCHING	Marches forward (arrow line,	Arrow line, len 300, w	None (march)	atkMul × base on contact + knockback	Rotat cader

	len 300, w 80) — pushes player (knockback) + damages on contact	80, t 180f			
A2 ARTILLERY	Cannons fire — 3 cannonballs (arrow, r 50, spd 0.35) arc at player's position, staggered 30f	3 arrow r 50, t 180f	3 cannonballs (arrow)	atkMul × base per cannonball	Rotat cader
A3 IRON WALL	Becomes invulnerable 3s — shield (arrow outline); player can't damage it, but Dreadnought can't attack either (holds position)	Arrow shield, t 180f, dur 180f	None (invulnerable)	None (invulnerable)	Rotat cader wall
SIG SIEGE	Holds position (4s) and fires a 5-shot cone every 0.5s in a FIXED direction (locked at fire time)	Arrow cone, t 180f, dur 240f	5-shot cones every 30f	atkMul × base per shot	Rotat cader siege 240f

STARWEAVER (void, star, atkMul 1.2)

Attack	Mechanic	Telegraph	Projectile	Damage
A1 STARLIGHT WEAVE	Stars weave a pattern — 8 stars (star) spawn in a weaving line	Star telegraphs along weave path, t 180f	8 weaving stars	atkMul × base per star

	(sine path) toward player			
A2 CONSTELLATION	Stars connect into a shape — 5 stars form a constellation (star pattern) at player's position, then the lines between them damage (burn)	Star constellation (5 stars + lines), t 180f	None (constellation burns)	atkMul × base if on a line at burn
A3 NEBULA	Cloud (star, r 120) hides the boss (invisible inside, 3s) + damages (0.5×/tick) if player enters	Star cloud r 120, t 180f, dur 180f	None (cloud)	0.5× base per tick × atkMul
SIG SUPERNOVA	Collapses into a star, then explodes (star ring r 250, 1s) + pulls player toward center	Star r 100 → star ring r 250, t 180f	Star ring	atkMul × base (explosion) + pull

BANANA KING (fire, quatrefoil, atkMul 1.2)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Th
A1 ROYAL DECREE	Decrees — buffs self (damage +20% for 5s) + quatrefoil ring (8 shots)	Quatrefoil ring r 120, t 180f	8 quatrefoil shots	atkMul × base per shot; buff +20% 300f	Rotation cadence; buff 300f	Th — em
A2 GOLDEN PEEL	Peels (quatrefoil, r 40) scatter — 5	5 quatrefoil peels r 40, t 180f	Peels (quatrefoil)	atkMul × base + slip/	Rotation cadence	Th pe sl

	peels land around player; stepping on one slips player (knockback + slow 1s) + damage			slow 60f		
A3 CORONATION	Crowns itself — gains a shield (blocks 1 hit) for 5s; while shielded, fires a quatrefoil fan (5 shots)	Quatrefoil crown r 100, t 180f, shield 300f	5 quatrefoil shots	atkMul × base per shot	Rotation cadence; shield 300f	Th co — is cr sh
SIG BANANA SPLIT	Splits into two (two quatrefoil Kings, each 50% HP) — both attack (fans) for 10s, then recombine (HP = sum)	Quatrefoil split, t 180f, dur 600f	Fans from both (5 shots each)	atkMul × base per shot	Rotation cadence; split 600f	Ba sp on be tw (r CR TO

PIP & POP (duo: Pip fire cross atkMul 1.0; Pop void ellipse atkMul 1.1)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme
A1 PEA FAN + MIRROR	Pip fires a fan (6, cross, green) — a distraction; Pop mirrors (blinks to player's side + fan 4, ellipse, pink)	Cross fan + ellipse fan, t 180f	6 cross + 4 ellipse	atkMul × base per shot (Pip 1.0 / Pop 1.1)	Rotation cadence	Pip distr Pop strik from side

A2 BULLY CHARGE + SQUABBLE	Pop charges (dash to player, 70f); Pip squabbles (ring 8, cross + fan 4, ellipse) from the other side	Ellipse dash + cross ring, t 180f	8 cross ring + 4 ellipse fan	atkMul × base per shot	Rotation cadence	Pop charg Pip squab — duo attac from sides
A3 SWAP	Pip & Pop swap positions (blink) — sprites swap for 3s, can't tell which is which; then both fire (Pip fan 5 cross + Pop fan 5 ellipse)	Cross + ellipse swap telegraphs, t 180f	5 cross + 5 ellipse	atkMul × base per shot	Rotation cadence; swap 180f	The s which which
SIG BETRAYAL	A "gift" appears (pickup, cross + ellipse) — if player takes it, it explodes (2 hits) + Pip blinks behind player and fires a fan (5, cross), Pop fires a ring (8, ellipse)	Gift telegraph (looks like a pickup), t 180f, then explosion + fans	Gift explosion + 5 cross fan + 8 ellipse ring	atkMul × base × 2 (gift) + per shot	Rotation cadence	Betra the g is a and t strik from behin (fold Pop's into swap)

ORACLE (lightning, heptagon, atkMul 1.15)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	T l
A1 REWIND	Player's position rewinds 2s	Heptagon r 60 at rewind	None (rewind)	atkMul × base if destination	Rotation cadence	T r y

	(moved back to where they were 2s ago); if rewind destination is inside a gear zone, damage	destination, t 180f		in a gear zone (else none — rewind itself is the disruption)		b w w
A2 TIME STOP	Player frozen (can't move, 1.5s) while gears (5, heptagon) fly at them	Heptagon freeze zone r 100, t 180f, freeze 90f	5 heptagon gears	atkMul × base per gear	Rotation cadence; freeze 90f	T s g
A3 GEAR CRUSH	Gears (heptagon, r 80 each) converge on player from 4 sides, then crush (damage if player still in center)	4 heptagon gears, t 180f, converge	None (converging gears)	atkMul × base on crush	Rotation cadence	T c
SIG CLOCK HANDS	Rotating beams (2 heptagon beams, len 420, r 30) sweep around Oracle (360° over 4s) — player must dodge between them	2 heptagon beams, t 180f, rotate 240f	2 rotating beams	atkMul × base per beam hit	Rotation cadence; sweep 240f	T h s t i a (C L f f i S

SUGAR WITCH (fire, circle, atkMul 1.10)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme link
A1 SYRUP POOL	Syrup pool (circle, r	Circle pool r 110, t	None (pool)	0.5× base		The s pool

	110) at player — sticky (0.5× speed) + damages over time (0.5×/tick, 1 tick/s, 4s)	180f, dur 240f		per tick × atkMul	Rotation cadence; pool 240f	stick burni
A2 CANDY SHRAPNEL	Candy (circle, r 60) flies at player, then explodes into shards (8, circle, spd 0.3) on impact or after 1s	Circle candy r 60, t 180f	1 candy + 8 shards	atkMul × base per shard	Rotation cadence; candy 60f fuse	The c explo into shard
A3 GUMDROP RAIN	Gumdrops (6, circle) rain from above (staggered) — each sticks to player on hit (slow stack, 1s each, up to 3 stacks) + damage	6 circle r 30, t 180f	6 gumdrops	atkMul × base per gumdrop + slow stack 60f each	Rotation cadence	Gumdr stick you
SIG SUGAR STORM	Meteors (5, circle, pink) rain down, each leaving a sticky pool (circle, r 60, 3s) that slows + damages (0.5×/tick)	5 circle meteor telegraphs, t 180f	5 meteors + 5 pools	atkMul × base per meteor + 0.5× per tick per pool	Rotation cadence; pools 180f	The s storm meteo that leave stick pools (keep SUGAR STORM upgra meteo leave pools

THORN REGENT (physical, pentagon, atkMul 1.15)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme lin
A1 THORN MAZE	Walls of thorns (4 pentagon walls forming a maze) block movement 5s — player must navigate; thorn walls damage on contact	4 pentagon walls, t 180f, dur 300f	None (walls)	atkMul × base on contact	Rotation cadence; maze 300f	The maze thorns block you (moves THORN MAZ from signature to regular)
A2 VINE GRAB	Vines (pentagon, r 30) shoot from Regent and grab player — if they connect, player is pulled toward Regent (1.0px/f, 1s) + damage	Pentagon r 70, t 150f	Vines (pentagon)	atkMul × base on grab + pull 60f	Rotation cadence	The vines pull you in
A3 BRAMBLE BURST	Thorns explode outward (pentagon ring, 12 shots, spd 0.3) from Regent	Pentagon ring r 150, t 180f	12 pentagon thorns	atkMul × base per thorn	Rotation cadence	The brambles burst
SIG ROOT SPIKE	Spikes erupt from the ground under player — 5 spikes	5 pentagon r 50, t 120f (fast)	Spikes (pentagon)	atkMul × base per spike	Rotation cadence	The roots spike — the groun attacks (replaces

	(pentagon, r 50) erupt in sequence (staggered 20f) at player's position (fast telegraph — must move)					THORN MAZ signature

LEVIATHAN (void, diamond, atkMul 1.20)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	The
A1 TIDAL SURGE	Tidal wave (roamer, r 80, t 400, vx 1.1) sweeps horizontally from the arena edge (direction based on player position) — dodge by moving UP or DOWN	Horizontal warning band (telegraph, t 150f, matching the wave's hitbox) + wave (roamer)	Wave (roamer, r 80)	atkMul × base if touched	Rotation cadence; wave 400f	The dod mov ver (ky kee lik
A2 WHIRLPOOL	Pull telegraph (circle, r 140, t 150, type pull) — pulls the player toward the center	Circle pull zone r 140, t 150f	None (pull)	atkMul × base on pull	Rotation cadence	The whi — p you (ky kee coo
A3 INK CLOUD	Ink cloud (diamond, r 150) blinds player (screen darkens, 2s) + hides Leviathan (invisible	Diamond cloud r 150, t 180f, dur 120f	None (cloud)	0.5× base per tick × atkMul	Rotation cadence; cloud 120f	The clo dar hid (ky rep Aby Spi coo

	inside) + damages (0.5×/tick) if player enters					
SIG CRUSHING DEPTHS	Crush telegraph (diamond, r 160, t 180, type crush) — crushes the player (big damage)	Diamond crush zone r 160, t 180f	None (crush)	atkMul × base (big damage)	Rotation cadence	The cru dep the cru you kee coo

CRYSTAL WARDEN (ice, beam, atkMul 1.20)

Attack	Mechanic	Telegraph	Projectile	Damage	Cooldown/ Duration	Theme link
A1 CRYSTAL SHARDS	Shards (8, beam-shaped) fly at player and REFLECT (bounce once off walls/ arena edges)	8 beam shards, t 180f	8 reflecting shards	atkMul × base per shard (2 possible hits: direct + bounce)	Rotation cadence	Shards reflect they bounce
A2 PRISM BEAM	Beam (len 420, r 30) fires, then SPLITS into 3 beams (±0.35 rad) at the midpoint	Beam + split, t 150f	1 beam → 3 beams	atkMul × base per beam	Rotation cadence	The pris splits the beam
A3 ICE SPLIT	Warden splits into 2 (beam-shaped, each 50% HP) for	Beam split, t 180f, dur 300f	Fans from both (4 shots each)	atkMul × base per shot	Rotation cadence; split 300f	The crystal splits — one becomes two (replace

	5s — both fire (fans), then recombine (HP = sum)					current icesplit telegrap with a true split)
SIG ICE PRISON	Crystal cage (beam-shaped walls, square cage r 100) forms around player 3s — can move inside but not leave; cage then shatters (damage if still inside) + shards (8) fly out	Beam cage r 100, t 180f, dur 180f	8 shards on shatter	atkMul × base (shatter) + per shard	Rotation cadence; cage 180f	The prison — trapped in crystal, then it shatters (keeps ICE PRISON, upgraded cage shatters into shards)

Attacks that could NOT be made pure-damage theme-specific (and why)

1. **Conductor A3 CRESCENDO** — A buff (accelerates the next 3 attacks), no direct damage. Theme-specific (the music gets faster) but it is a setup attack, not a hit. Kept because the orchestra-speeds-up mechanic only works as a buff.
2. **Librarian A1 SHUSH** — A lock (silence), no damage. Theme-specific (silence in the library) but it locks skills instead of dealing damage. Kept because the silence mechanic is the library's identity.
3. **Dreadnought A3 IRON WALL** — A defensive buff (invulnerable 3s, no damage). Theme-specific (the iron wall) but it is a turtle, not a hit. Kept because the war-machine-turtles-up mechanic needs a non-damage window to be fair (player gets a breather while the boss can't attack either).
4. **Oracle A1 REWIND** — The rewind itself deals no damage (damage only if the rewind destination is inside a gear

zone). Theme-specific (time rewinds) but the disruption is positional, not damage. Kept because a forced rewind is the time mechanic; damage comes from the gear-zone interaction.
5. **Starweaver A3 NEBULA** — Hide + DoT cloud (0.5×/tick). Theme-specific (the nebula hides) but the boss-invisibility needs the cloud to be the threat. Kept as a hide + DoT (not a direct hit).

Implementation notes (for the implementer, after cohorts)

1. Reuse existing primitives where possible: bFan/bRing/bMeteor/bSlow/bCracks/bBurn (projectiles/zones), roamers (tidal wave, boulder, crown, tornado), telegraphs (zones/beams/pull/silence/prison/crush), bSummon (wolves/bees), bBlink (blinks/swaps), z.dashT (dashes/pounces/charges), z.spinT/spinMode (spin attacks — replace with theme mechanics where noted). New primitives needed: rewind (store player position history, 2s), converging gears (4 telegraphs moving inward), reflecting shards (bounce once off arena edges), split-boss (two entities, recombine HP = sum), gift pickup (fake pickup that explodes), debt DoT (pauses while moving), pull-toward-boss (directional pull), rotating beams (2 beams sweeping 360° over 240f), honey pool burst, syrup pool DoT, ink-cloud blind (screen darken), mirror reflect (fan bounces off mirror walls).
2. The OHK attack (waves 30–31) is unchanged (it is a separate system, not part of the 4-attack rotation).
3. P1 rotation = [seq0, seq1, signature]; NG rotation = [seq0, seq1, seq2, signature] — unchanged (the new 4-attack sets slot into the same rotation slots).
4. ATK_TELL shout names must be updated to the new attack names (e.g., REWIND!, TIME STOP!, GEAR CRUSH!, CLOCK HANDS!, SYRUP POOL!, CANDY SHRAPNEL!, GUMDROP RAIN!, SUGAR STORM!, THORN MAZE!, VINE GRAB!, BRAMBLE BURST!, ROOT SPIKE!, TIDAL SURGE!, WHIRLPOOL!, INK CLOUD!, CRUSHING DEPTHS!, CRYSTAL SHARDS!, PRISM BEAM!, ICE SPLIT!, ICE PRISON!, plus the 15 originals' new names).
5. Boss music / sprites / loot / stats are NOT changed by this spec (mechanics only).